
gen4 Devices Feature Description

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ACQUISITIONCONTROL

Category for the acquisition and trigger control features.

1.1 AcquisitionMode

Name	AcquisitionMode
Type	Enumeration
Visibility	Beginner
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	Continuous
	SingleFrame

Sets the acquisition mode of the device. It defines mainly the number of frames to capture during an acquisition and the way the acquisition stops.

- Continuous: Frames are captured continuously until stopped with the AcquisitionStop command.
- SingleFrame: One frame is captured.

1.2 AcquisitionStart

Name	AcquisitionStart
Type	Command
Visibility	Beginner
Writable in Acquisition Mode	False
AccessMode	RW

Starts the acquisition of the device. The number of frames captured is specified by AcquisitionMode.

1.3 AcquisitionStop

Name	AcquisitionStop
Type	Command
Visibility	Beginner
AccessMode	RW

Stops the acquisition of the device at the end of the current frame. It is mainly used when AcquisitionMode is Continuous but can be used in any acquisition mode.

1.4 AcquisitionStatusSelector

Name	AcquisitionStatusSelector
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	FrameTriggerWait
	RecordingTriggerWait
	FrameActive
	RecordingActive

Select the internal acquisition signal to read using the AcquisitionStatus feature.

- FrameTriggerWait: Device is currently waiting for a frame start trigger.
- RecordingTriggerWait: Device is currently waiting for a recording start trigger.
- FrameActive: Device is currently doing the capture and processing of a frame.
- RecordingActive: Device is currently recording data.

1.5 AcquisitionStatus

Name	AcquisitionStatus [AcquisitionStatusSelector]
Type	Boolean
Visibility	Expert
AccessMode	RO

Reads the state of the internal acquisition signal using the AcquisitionStatusSelector.

1.6 AcquisitionBurstFrameCount

Name	AcquisitionBurstFrameCount
Type	Integer
Visibility	Beginner
Operation Mode	LockInCam
Writable in Acquisition Mode	False
AccessMode	RW
Inc	2
Min	depends on feature <i>AcquisitionBurstFrameCountMin</i>
Max	depends on feature <i>AcquisitionBurstFrameCountMax</i>

Number of frames to acquire for each FrameBurstStart trigger. This feature is used only if the FrameBurstStart trigger is enabled and the FrameBurstEnd trigger is disabled. Note that the total number of frames captured is also conditioned by AcquisitionFrameCount if AcquisitionMode is MultiFrame and ignored if AcquisitionMode is Single.

1.7 TriggerSelector

Name	TriggerSelector
Type	Enumeration
Visibility	Beginner
AccessMode	RW
EnumEntry	
	AcquisitionStart
	FrameStart
	FrameBurstStart
	RecordingStart
	Reference

Select the type of trigger to configure.

- AcquisitionStart: (**Deprecated!**) Deprecated: Use FrameStart. The correct implementation according to the GenICam standard of AcquisitionStart was never released on heliInspect. Selects a trigger that starts the Acquisition of one or many frames according to AcquisitionMode.
- FrameStart: Selects a trigger starting the capture of one frame (3D surface).
- FrameBurstStart: This trigger is not fully implemented now (Please don't use it). Select a trigger starting the capture of the bursts of frames in an acquisition. AcquisitionBurstFrameCount controls the length of each burst unless a FrameBurstEnd trigger is active. The total number of frames captured is also conditioned by AcquisitionFrameCount if AcquisitionMode is MultiFrame.
- RecordingStart: Select the trigger source to start the data recording. By default, the recording is started and controlled by the motion. In Auto mode, the recording starts immediately after the acquisition is started.
- Reference: Select the trigger controlling the demodulation frequency.

1.8 TriggerMode

Name	TriggerMode [TriggerSelector]
Type	Enumeration
Visibility	Beginner
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	Off
	On

Controls if the selected trigger is active.

- Off: Disables the selected trigger.
- On: Enable the selected trigger.

1.9 TriggerSoftware

Name	TriggerSoftware [TriggerSelector]
Type	Command
Visibility	Beginner
AccessMode	WO

Generates an internal trigger. TriggerSource must be set to Software.

1.10 TriggerSource

Name	TriggerSource [TriggerSelector]
Type	Enumeration
Visibility	Beginner
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	Software
	Auto
	Stage
	RTIO4
	DIO1
	DIO2
	DIO3
	DIO4
	DIO5
	DIO6
	DIO7
	DIO8
	DIO10
	DIO11
	DIO12
	DIO13

continues on next page

Table 1.1 – continued from previous page

DIO14
DIO15
DIO16
DIO17
FI2
FI3
RTIO6
RTIO7
FieldbusBool0
FieldbusBool1
FieldbusBool2
FieldbusBool3
FieldbusBool4
FieldbusBool5
FieldbusBool6
FieldbusBool7
EncoderCamera

Specifies the internal signal or physical input Line to use as the trigger source. The selected trigger must have its TriggerMode set to On.

- Software: Specifies that the trigger source will be generated by software using the TriggerSoftware command.
- Auto: (**Deprecated!!**) Deprecated: According to the GenICam specification this functionality is equal to TriggerMode=Off. Specifies that the trigger source will be generated automatically as fast as possible.
- Stage: Specifies that the trigger source for recording start is generated by the stage. Note: Only available for RecordingStart Trigger.
- RTIO4: Specifies that the trigger will be generated by input RTIO4.
- DIO1: Specifies that the trigger will be generated by input DIO1.
- DIO2: Specifies that the trigger will be generated by input DIO2.
- DIO3: Specifies that the trigger will be generated by input DIO3.
- DIO4: Specifies that the trigger will be generated by input DIO4.
- DIO5: Specifies that the trigger will be generated by input DIO5.
- DIO6: Specifies that the trigger will be generated by input DIO6.
- DIO7: Specifies that the trigger will be generated by input DIO7.
- DIO8: Specifies that the trigger will be generated by input DIO8.
- DIO10: Specifies that the trigger will be generated by input DIO10.
- DIO11: Specifies that the trigger will be generated by input DIO11.
- DIO12: Specifies that the trigger will be generated by input DIO12.
- DIO13: Specifies that the trigger will be generated by input DIO13.
- DIO14: Specifies that the trigger will be generated by input DIO14.
- DIO15: Specifies that the trigger will be generated by input DIO15.
- DIO16: Specifies that the trigger will be generated by input DIO16.
- DIO17: Specifies that the trigger will be generated by input DIO17.
- FI2: Specifies that the trigger will be generated by input FI2. Note: Only available for RecordingStart and Reference Trigger.

- FI3: Specifies that the trigger will be generated by input FI3. Note: Only available for RecordingStart and Reference Trigger.
- RTIO6: Specifies that the trigger will be generated by input RTIO6. Note: Only available for RecordingStart and Reference Trigger.
- RTIO7: Specifies that the trigger will be generated by input RTIO7. Note: Only available for RecordingStart and Reference Trigger.
- FieldbusBool0: Specifies that the trigger will be generated by the FieldbusBool slot 0.
- FieldbusBool1: Specifies that the trigger will be generated by the FieldbusBool slot 1.
- FieldbusBool2: Specifies that the trigger will be generated by the FieldbusBool slot 2.
- FieldbusBool3: Specifies that the trigger will be generated by the FieldbusBool slot 3.
- FieldbusBool4: Specifies that the trigger will be generated by the FieldbusBool slot 4.
- FieldbusBool5: Specifies that the trigger will be generated by the FieldbusBool slot 5.
- FieldbusBool6: Specifies that the trigger will be generated by the FieldbusBool slot 6.
- FieldbusBool7: Specifies that the trigger will be generated by the FieldbusBool slot 7.
- EncoderCamera: Specifies that the trigger will be generated by the camera encoder. Note: Only available with internal motion control system or external stage.

1.11 TriggerActivation

Name	TriggerActivation [TriggerSelector]
Type	Enumeration
Visibility	Beginner
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	RisingEdge
	FallingEdge
	AnyEdge

Specifies the activation mode of the trigger.

- RisingEdge: Specifies that the trigger is considered valid on the rising edge of the source signal.
- FallingEdge: Specifies that the trigger is considered valid on the falling edge of the source signal.
- AnyEdge: Specifies that the trigger is considered valid on the rising or falling edge of the source signal.

1.12 TriggerOverlap

Name	TriggerOverlap [TriggerSelector]
Type	Enumeration
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	Off
	PreviousFrame

Specifies the type trigger overlap permitted with the previous frame. This defines when a valid trigger will be accepted (or latched) for a new frame.

- Off: No trigger overlap is permitted.
- PreviousFrame: Trigger is accepted (latched) at any time during the capture of the previous frame.

1.13 TriggerDelay

Name	TriggerDelay [TriggerSelector]
Type	Float
Visibility	Expert
AccessMode	RW
Unit	us
Min	0.0
Max	489.6
Inc	0.015

Specifies the delay in microseconds (us) to apply after the trigger reception before activating it.

1.14 TriggerDivider

Name	TriggerDivider [TriggerSelector]
Type	Integer
Visibility	Expert
AccessMode	RW
Min	1
Max	depends on feature <i>TriggerDividerMax</i>

Specifies a division factor for the incoming trigger pulses.

1.15 TriggerMultiplier

Name	TriggerMultiplier [TriggerSelector]
Type	Integer
Visibility	Expert
AccessMode	RW
Min	1
Max	depends on feature <i>TriggerMultiplierMax</i>

Specifies a multiplication factor for the incoming trigger pulses. It is generally used in conjunction with TriggerDivider to control the ratio of triggers that are accepted.

CHUNKDATACONTROL

Category that contains the Chunk Data control features.

2.1 ChunkModeActive

Name	ChunkModeActive
Type	Boolean
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW

Activates the inclusion of Chunk data in the payload of the image.

2.2 ChunkSelector

Name	ChunkSelector
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	
	PartCount
	PartFixpointScaling
	PartFrameNo
	PartType
	ZTagCount
	ZTagValue
	Scan3dDistanceUnit
	Scan3dOutputMode
	Scan3dCoordinateScale
	Scan3dCoordinateOffset
	ProcessingString
	ConfigString
	Timestamp
	FrameID

Select which Chunk to enable or control.

- PartCount: Enable ChunkPartCount feature in chunk data.
- PartFixpointScaling: Enable ChunkPartFixpointScaling feature in chunk data.

- PartFrameNo: Enable ChunkPartFrameNo feature in chunk data.
- PartType: Enable ChunkPartType feature in chunk data.
- ZTagCount: Enable ChunkZTagCount feature in chunk data.
- ZTagValue: Enable ChunkZTagValue feature in chunk data.
- Scan3dDistanceUnit: Enable ChunkScan3dDistanceUnit feature in chunk data.
- Scan3dOutputMode: Enable ChunkScan3dOutputMode feature in chunk data.
- Scan3dCoordinateScale: Enable ChunkScan3dCoordinateScale feature in chunk data.
- Scan3dCoordinateOffset: Enable ChunkScan3dCoordinateOffset feature in chunk data.
- ProcessingString: Enable ChunkProcessingString feature in chunk data.
- ConfigString: Enable ChunkConfigString feature in chunk data.
- Timestamp: Enable Timestamp feature in chunk data.
- FrameID: Enable FrameID feature in chunk data.

2.3 ChunkEnable

Name	ChunkEnable [ChunkSelector]
Type	Boolean
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW

Enables the inclusion of the selected Chunk data in the payload of the image.

2.4 ChunkPartCount

Name	ChunkPartCount
Type	IntReg
Visibility	Expert
AccessMode	RO

Total number of parts included in this buffer.

2.5 ChunkPartSelector

Name	ChunkPartSelector
Type	Integer
Visibility	Expert
Min	0
Inc	1
Max	depends on feature <i>ChunkPartCount</i>

Select the part number from which the meta information is returned.

2.6 ChunkPartFixpointScaling

Name	ChunkPartFixpointScaling
Type	FloatReg
Visibility	Expert
AccessMode	RO

Correction factor for fixpoint scaling of selected part.

2.7 ChunkPartFrameNo

Name	ChunkPartFrameNo
Type	IntReg
Visibility	Expert
AccessMode	RO

Frame number or number of subcomponent of selected part.

2.8 ChunkPartType

Name	ChunkPartType
Type	Enumeration
Visibility	Expert
AccessMode	RO
EnumEntry	
	Unknown
	Surface
	Amplitude
	Phase
	Confidence
	pZglobal
	pZlocal
	pZenvelope
	pP
	unwrap_group
	Zenvelope
	I
	Q

Enumeration gives the channel type to which the part selected by ChunkPartSelector belongs.

- Unknown: Unknown part type or type not available now.
- Surface: (enabled by component *Range* or *Intensity*) This part contains range information. The numeric value represents range in units specified as the product of *ChunkPartFixpointScaling* times *ChunkScan3dDistanceUnit*.
- Amplitude: (enabled by component *Reflectance*) This part contains amplitude information. The numeric value represents estimated ADC signal amplitude at the peak of the signal, expressed in units of *ChunkPart-FixpointScaling* ADC counts.

- Phase: (enabled by component *Phase*) This part contains phase information. It represents the demodulation phase at the peak of the signal, expressed in units of *ChunkPartFixpointScaling* radians.
- Confidence: (enabled by component *Confidence*) This part contains confidence information. The numeric value represents the Bayesian posterior probability that the extracted surface information does represent a real surface (as opposed to random noise), as a fraction in the range 0 to 1.
- pZglobal: (enabled by component *RangePrecision* and configured by *PrecisionScale*) This part contains global range precision or uncertainty information. If *PrecisionScale* is set to *InverseStandardDeviation*, the output is precision (which is the inverse of uncertainty). The numeric values represents either one sigma of uncertainty in units specified as the product of *ChunkPartFixpointScaling* times *ChunkScan3dDistanceUnit*, or the inverse of that.
- pZlocal: (enabled by component *RangePrecision* and configured by *PrecisionScale*) This part contains local range precision or uncertainty information, valid for the comparison between pixels that are within the same phase-unwrapping region/group (see also channel *unwrap-group*). If *PrecisionScale* is set to *InverseStandardDeviation*, the output is precision (which is the inverse of uncertainty). The numeric values represents either one sigma of uncertainty in units specified as the product of *ChunkPartFixpointScaling* times *ChunkScan3dDistanceUnit*, or the inverse of that.
- pZenvelope: (enabled by component *AdditionalProcessingChannels*) This part contains the equivalent of *pZ* of the intermediate (envelope-based) centre of mass algorithm, when followed by phase-unwrapping.
- pP: (enabled by component *AdditionalProcessingChannels* and configured by *PrecisionScale*) This part contains the precision or uncertainty information of the phase component. If *PrecisionScale* is set to *InverseStandardDeviation*, the output is precision (which is the inverse of uncertainty). The numeric values represents either one sigma of uncertainty in units of radians, or the inverse of that.
- unwrap_group: (enabled by component *RangePrecision*) This part associates each pixel with a phase-unwrapping group, identified by the numerical index value. The special value 0 is assigned to pixels belonging to very small phase-unwrapping group, typically low confidence pixels. Pixels within the same group are connected through successful phase-unwrapping, and therefore the relative accuracy is better than between independent groups. Then, the relative accuracy should be determined from *sZ-local/pZ-local* rather than *sZ/pZ*.
- Zenvelope: (enabled by component *AdditionalProcessingChannels*) This part contains the equivalent of *Z* of the intermediate (envelope-based) centre of mass algorithm, when followed by phase-unwrapping.
- I: (enabled by component *Intensity*) in-phase information of raw data. Only available in 'Scan3dExtractionMethod' = rawIQ
- Q: (enabled by component *Intensity*) Quadrature information of raw data. Only available in 'Scan3dExtractionMethod' = rawIQ

2.9 ChunkZTagCount

Name	ChunkZTagCount
Type	IntReg
Visibility	Expert
AccessMode	RO

Return the maximum number of zTags included in the measurement.

2.10 ChunkZTagSelector

Name	ChunkZTagSelector
Type	Integer
Visibility	Expert
Min	0
Inc	1
Max	depends on feature <i>ChunkZTagSelectorMax</i>

Index of zTag to read.

2.11 ChunkZTagValue

Name	ChunkZTagValue
Type	IntReg
Visibility	Expert
AccessMode	RO

zTag value from selected position (ChunkZTagSelector).

2.12 ChunkScan3dDistanceUnit

Name	ChunkScan3dDistanceUnit
Type	Enumeration
Visibility	Expert
AccessMode	RO
EnumEntry	Frames
	nm
	um
	mm

Returns the Distance Unit of the payload image (Range and Scaled intensity).

- Frames: Frame based surface value.
- nm: Distance values are in nanometer units.
- um: Distance values are in micrometer units.
- mm: Distance values are in millimeter units.

2.13 ChunkScan3dOutputMode

Name	ChunkScan3dOutputMode
Type	Enumeration
Visibility	Expert
AccessMode	RO
EnumEntry	
	CalibratedC
	RectifiedC

Returns the Calibrated Mode of the payload image.

- CalibratedC: Calibrated 2.5D Depth map. The distance data is expressed in the chosen distance unit. The data is a 2.5D range map. No information on X-Y axes available.
- RectifiedC: Rectified 2.5D depth map. The distance data has been rectified to a uniform sampling pattern in the X and Y direction. The data is a 2.5D range map only.

2.14 ChunkScan3dCoordinateSelector

Name	ChunkScan3dCoordinateSelector
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	
	CoordinateA
	CoordinateB
	CoordinateC

Select which Coordinate to retrieve data from.

- CoordinateA: The first (X or Theta) coordinate.
- CoordinateB: The second (Y or Phi) coordinate.
- CoordinateC: The third (Z or Rho) coordinate.

2.15 ChunkScan3dCoordinateScale

Name	ChunkScan3dCoordinateScale [ChunkScan3dCoordinateSelector]
Type	FloatReg
Visibility	Expert
AccessMode	RO

Returns the Scale for the selected coordinate axis of the image included in the payload.

2.16 ChunkScan3dCoordinateOffset

Name	ChunkScan3dCoordinateOffset [ChunkScan3dCoordinateSelector]
Type	FloatReg
Visibility	Expert
AccessMode	RO

Returns the Offset for the selected coordinate axis of the image included in the payload.

2.17 ChunkProcessingString

Name	ChunkProcessingString
Type	StringReg
Visibility	Guru
AccessMode	RO
Length	65535

JSON formatted string containing processing information, log messages and timing information.

2.18 ChunkConfigString

Name	ChunkConfigString
Type	StringReg
Visibility	Guru
AccessMode	RO
Length	65535

JSON formatted string containing the whole camera settings and meta information of the measurement.

2.19 ChunkPixelFormat

Name	ChunkPixelFormat
Type	Enumeration
Visibility	Expert
AccessMode	RO
EnumEntry	
	Mono8
	Mono16
	Coord3D_C32f
	Confidence32f

Returns the PixelFormat of the selected Part (ChunkPartSelector).

- Mono8: Monochrome 8-bit
- Mono16: Monochrome 16-bit

- Coord3D_C32f: 3D coordinate C 32-bit floating point
- Confidence32f: Confidence 32-bit floating point

2.20 ChunkFrameID

Name	ChunkFrameID
Type	IntReg
Visibility	Expert
AccessMode	RO

Returns the unique Identifier of the frame (or image) included in the payload. It starts at 0 and keeps incrementing by one for each frame up to a maximum, then it wraps to 0 again. Only a device reboot or power-cycle resets the counter.

2.21 ChunkTimestamp

Name	ChunkTimestamp
Type	IntReg
Visibility	Expert
AccessMode	RO
Unit	ns

Returns the Timestamp of the image included in the payload at the time of the FrameStart internal event.

COUNTERANDTIMERCONTROL

Category that contains the Counter and Timer control features.

3.1 CounterSelector

Name	CounterSelector
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	
	Counter0
	Counter1

Select which counter to configure.

- Counter0: Select the counter 0
- Counter1: Select the counter 1

3.2 CounterEventSource

Name	CounterEventSource [CounterSelector]
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	
	Off
	AcquisitionTrigger
	AcquisitionTriggerMissed
	FrameTrigger
	FrameTriggerMissed
	RTIO4
	DIO1
	DIO2
	DIO3
	DIO4
	DIO5
	DIO6
	DIO7
	DIO8

continues on next page

Table 3.1 – continued from previous page

DIO10
DIO11
DIO12
DIO13
DIO14
DIO15
DIO16
DIO17
RecordingTrigger
RecordingTriggerMissed
FieldbusBool0
FieldbusBool1
FieldbusBool2
FieldbusBool3
FieldbusBool4
FieldbusBool5
FieldbusBool6
FieldbusBool7

Select the events that will be the source to increment the counter.

- Off: Counter is stopped
- AcquisitionTrigger: Counts the number of AcquisitionStart Triggers.
- AcquisitionTriggerMissed: Counts the number of missed AcquisitionStart Triggers.
- FrameTrigger: Counts the number of FrameStart Triggers.
- FrameTriggerMissed: Counts the number of missed FrameStart Triggers.
- RTIO4: Counts the number of transitions on I/O Line RTIO4.
- DIO1: Counts the number of transitions on I/O Line DIO1.
- DIO2: Counts the number of transitions on I/O Line DIO2.
- DIO3: Counts the number of transitions on I/O Line DIO3.
- DIO4: Counts the number of transitions on I/O Line DIO4.
- DIO5: Counts the number of transitions on I/O Line DIO5.
- DIO6: Counts the number of transitions on I/O Line DIO6.
- DIO7: Counts the number of transitions on I/O Line DIO7.
- DIO8: Counts the number of transitions on I/O Line DIO8.
- DIO10: Counts the number of transitions on I/O Line DIO10.
- DIO11: Counts the number of transitions on I/O Line DIO11.
- DIO12: Counts the number of transitions on I/O Line DIO12.
- DIO13: Counts the number of transitions on I/O Line DIO13.
- DIO14: Counts the number of transitions on I/O Line DIO14.
- DIO15: Counts the number of transitions on I/O Line DIO15.
- DIO16: Counts the number of transitions on I/O Line DIO16.
- DIO17: Counts the number of transitions on I/O Line DIO17.
- RecordingTrigger: Counts the number of RecordingStart Triggers.
- RecordingTriggerMissed: Counts the number of missed RecordingStart Triggers.

- FieldbusBool0: Counts the number of transitions on FieldbusBool slot 0.
- FieldbusBool1: Counts the number of transitions on FieldbusBool slot 1.
- FieldbusBool2: Counts the number of transitions on FieldbusBool slot 2.
- FieldbusBool3: Counts the number of transitions on FieldbusBool slot 3.
- FieldbusBool4: Counts the number of transitions on FieldbusBool slot 4.
- FieldbusBool5: Counts the number of transitions on FieldbusBool slot 5.
- FieldbusBool6: Counts the number of transitions on FieldbusBool slot 6.
- FieldbusBool7: Counts the number of transitions on FieldbusBool slot 7.

3.3 CounterEventActivation

Name	CounterEventActivation [CounterSelector]
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	
	RisingEdge
	FallingEdge
	AnyEdge

Select the Activation mode of the CounterEventSource signal. Feature is used if CounterSource is set to physical signals (RTIOx, DIOx).

- RisingEdge: Counts on the Rising Edge of the selected signal.
- FallingEdge: Counts on the Falling Edge of the selected signal.
- AnyEdge: Counts on the Rising or Falling Edge of the selected signal.

3.4 CounterReset

Name	CounterReset [CounterSelector]
Type	Command
Visibility	Expert
AccessMode	WO

Does a software reset of the selected counter.

3.5 CounterValue

Name	CounterValue [CounterSelector]
Type	IntReg
Visibility	Expert
AccessMode	RO

Reads the current value of the selected counter.

DACDEVICE

Low-Level DAC control features

DEVICECONTROL

Category for device information and control.

5.1 DeviceVendorName

Name	DeviceVendorName
Type	StringReg
Visibility	Beginner
AccessMode	RO
Length	16

Name of the manufacturer of the device.

5.2 DeviceModelName

Name	DeviceModelName
Type	StringReg
Visibility	Beginner
AccessMode	RO
Length	16

Model of the device.

5.3 DeviceFamilyName

Name	DeviceFamilyName
Type	StringReg
Visibility	Beginner
AccessMode	RO
Length	16

Identifier of the product family of the device.

5.4 DeviceFirmwareVersion

Name	DeviceFirmwareVersion
Type	StringReg
Visibility	Beginner
AccessMode	RO
Length	16

Version of the firmware in the device.

5.5 DeviceKernelVersion

Name	DeviceKernelVersion
Type	StringReg
Visibility	Guru
AccessMode	RO
Length	16

Linux kernel version executed on the device.

5.6 DeviceDriverVersion

Name	DeviceDriverVersion
Type	StringReg
Visibility	Guru
AccessMode	RO
Length	48

Linux kernel module version on the device.

5.7 DeviceFPGAFwVersion

Name	DeviceFPGAFwVersion
Type	IntReg
Visibility	Guru
AccessMode	RO

Version of the FPGA firmware in the device.

5.8 DeviceSerialNumber

Name	DeviceSerialNumber
Type	StringReg
Visibility	Expert
AccessMode	RO
Length	16

Device's serial number. This string is a unique identifier of the device.

5.9 DeviceTemperatureSelector

Name	DeviceTemperatureSelector
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	
	Mainboard
	Processor
	Optic

Select the location within the device, where the temperature will be measured.

- Mainboard: Temperature of the device mainboard.
- Processor: Temperature of the device processor.
- Optic: Temperature of the optics.

5.10 DeviceTemperature

Name	DeviceTemperature [DeviceTemperatureSelector]
Type	FloatReg
Visibility	Expert
AccessMode	RO
Unit	C

Device temperature in degrees Celsius (C). It is measured at the location selected by DeviceTemperatureSelector.

5.11 DeviceMonitorSelector

Name	DeviceMonitorSelector
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	
	VddVoltage
	VddCurrent
	VddPower
	UserVddVoltage
	UserVddCurrent
	UserVddPower
	VddDrv
	OpticTemperatureControlPower

Select the location within the device, where the monitoring information is read.

- VddVoltage: Vdd Voltage in [V].
- VddCurrent: Vdd Current in [A]
- VddPower: Vdd Power in [W]
- UserVddVoltage: UserVdd Voltage in [V]
- UserVddCurrent: UserVdd Current in [A]
- UserVddPower: UserVdd Power in [W]
- VddDrv: LED driver Voltage in [V]
- OpticTemperatureControlPower: Current Power in [W] used for the thermal control of the optics.

5.12 DeviceMonitorActualValue

Name	DeviceMonitorActualValue [DeviceMonitorSelector]
Type	FloatReg
Visibility	Expert
AccessMode	RO

Actual Value of selected monitoring feature. Units according to the selected monitoring feature.

5.13 DeviceMonitorMinimumValue

Name	DeviceMonitorMinimumValue [DeviceMonitorSelector]
Type	FloatReg
Visibility	Expert
AccessMode	RO

Minimum Value since powerup of selected monitoring feature. Units according to the selected monitoring feature.

5.14 DeviceMonitorMaximumValue

Name	DeviceMonitorMaximumValue [DeviceMonitorSelector]
Type	FloatReg
Visibility	Expert
AccessMode	RO

Maximum Value since powerup of selected monitoring feature. Units according to the selected monitoring feature.

5.15 Timestamp

Name	Timestamp
Type	IntReg
Visibility	Expert
AccessMode	RO
Unit	ns

Reports the current value of the device timestamp counter. The same timestamp counter is used for tagging images, chunk and event data.

5.16 DeviceReset

Name	DeviceReset
Type	Command
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	WO

Resets the device to its power up state.

5.17 TimestampReset

Name	TimestampReset
Type	Command
Visibility	Expert
AccessMode	WO

Resets the current value of the device timestamp counter. After executing this command, the timestamp counter restarts automatically.

DIGITAL I/O CONTROL

Category that contains the digital input and output control features.

6.1 LineSelector

Name	LineSelector
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	
	Line0
	Line1
	Line2
	Line3
	RTIO2
	RTIO3
	RTIO4
	DIO1
	DIO2
	DIO3
	DIO4
	DIO5
	DIO6
	DIO7
	DIO8
	DIO10
	DIO11
	DIO12
	DIO13
	DIO14
	DIO15
	DIO16
	DIO17
	RTIO6
	RTIO7
	FI2
	FI3
	SwCtrlVdd5

Select the physical line (or pin) of the external device connector or the virtual line of the Transport Layer to configure. When a Line is selected, all the other Line features will be applied to its associated I/O control block and will condition the resulting input or output signal.

- Line0: Digital IO 0: HPS Input 0, HpsHhprt1v8io0
- Line1: Digital IO 1: HPS Input 1, HpsHhprt1v8io1
- Line2: Digital IO 2: HPS Output 0, HpsHhprt1v8io2
- Line3: Digital IO 3: HPS Output 1, HpsHhprt1v8io3
- RTIO2: heliDriver D3 realtime output RTIO2 on RTIO or LIA Module
- RTIO3: heliDriver D3 realtime output RTIO3 on RTIO or LIA Module
- RTIO4: heliDriver D3 digital input RTIO4 on RTIO Module
- DIO1: heliDriver D3 digital input/output DIO1 on DIO or LIA Module
- DIO2: heliDriver D3 digital input/output DIO2 on DIO or LIA Module
- DIO3: heliDriver D3 digital input/output DIO3 on DIO or LIA Module
- DIO4: heliDriver D3 digital input/output DIO4 on DIO or LIA Module
- DIO5: heliDriver D3 digital input/output DIO5 on DIO or LIA Module
- DIO6: heliDriver D3 digital input/output DIO6 on DIO or LIA Module
- DIO7: heliDriver D3 digital input/output DIO7 on DIO or LIA Module
- DIO8: heliDriver D3 digital input/output DIO8 on DIO or LIA Module
- DIO10: heliDriver D3 digital input/output DIO10 on DIO or LIA Module
- DIO11: heliDriver D3 digital input/output DIO11 on DIO or LIA Module
- DIO12: heliDriver D3 digital input/output DIO12 on DIO or LIA Module
- DIO13: heliDriver D3 digital input/output DIO13 on DIO or LIA Module
- DIO14: heliDriver D3 digital input/output DIO14 on DIO or LIA Module
- DIO15: heliDriver D3 digital input/output DIO15 on DIO or LIA Module
- DIO16: heliDriver D3 digital input/output DIO16 on DIO or LIA Module
- DIO17: heliDriver D3 digital input/output DIO17 on DIO or LIA Module
- RTIO6: heliDriver D3 realtime input RTIO6 on RTIO or LIA Module
- RTIO7: heliDriver D3 realtime input RTIO7 on RTIO or LIA Module
- FI2: heliDriver D3 realtime input FI2 on RTIO or LIA Module
- FI3: heliDriver D3 realtime input FI3 on RTIO or LIA Module
- SwCtrlVdd5: heliDriver D3 5v power output on LIA Module

6.2 LineMode

Name	LineMode [LineSelector]
Type	Enumeration
Visibility	Expert
Writable in Acquisition Mode	True
AccessMode	RW
EnumEntry	
	Input
	Output

Controls if the physical Line is used to Input or Output a signal. Depending on the hardware, this configuration is read only or read/write. When a Line supports input and output mode, the default state is Input to avoid possible electrical contention.

- Input: The selected physical line is used to Input an electrical signal.
- Output: The selected physical line is used to Output an electrical signal.

6.3 LineType

Name	LineType [LineSelector]
Type	Enumeration
Visibility	Expert
AccessMode	RO
EnumEntry	Unknown
	HPS
	FPGA
	D3RTIO
	D3DIO
	D3LIA

Gives the hardware type and control source of the selected Line.

- Unknown: The line is controlled by the HPS.
- HPS: The line is controlled by the HPS.
- FPGA: The line is controlled by the FPGA. The functionality is static.
- D3RTIO: The line is controlled by the heliDriver D3 RTIO module.
- D3DIO: The line is controlled by the heliDriver D3 DIO module.
- D3LIA: The line is controlled by the heliDriver D3 LIA module.

6.4 LineInverter

Name	LineInverter [LineSelector]
Type	Boolean
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW

Controls the inversion of the signal of the selected input or output Line.

6.5 LineStatus

Name	LineStatus [LineSelector]
Type	Boolean
Visibility	Expert
AccessMode	RO

Returns the current status of the selected input or output Line.

6.6 LineStatusAll

Name	LineStatusAll
Type	IntReg
Visibility	Expert
AccessMode	RO

Returns the current status of all available Line signals.

6.7 LineSource

Name	LineSource [LineSelector]
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	
	Off
	UserOutput0
	UserOutput1
	UserOutput2
	UserOutput3
	UserOutput4
	UserOutput5
	UserOutput6
	UserOutput7
	FrameTriggerWait
	FrameTriggerMissed
	FrameActive
	RecordingTriggerWait
	RecordingTriggerMissed
	RecordingActive
	StreamTransferActive
	StageInMotion
	LockInReference

Selects which internal acquisition or I/O source signal to output on the selected Line. LineMode must be Output.

- Off: Selected Line is not an output or output is off.
- UserOutput0: Map Selected Line to UserOutput0.
- UserOutput1: Map Selected Line to UserOutput1.

- UserOutput2: Map Selected Line to UserOutput2.
- UserOutput3: Map Selected Line to UserOutput3.
- UserOutput4: Map Selected Line to UserOutput4.
- UserOutput5: Map Selected Line to UserOutput5.
- UserOutput6: Map Selected Line to UserOutput6.
- UserOutput7: Map Selected Line to UserOutput7.
- FrameTriggerWait: Device is currently waiting for a FrameStart trigger to start a measurement.
- FrameTriggerMissed: Device has missed a FrameStart trigger. Is reset when switching between acquisition and configuration mode.
- FrameActive: Device is currently doing a measurement.
- RecordingTriggerWait: Device is currently waiting for a recording start trigger.
- RecordingTriggerMissed: Device has missed a Recording trigger. Is reset when switching between acquisition and configuration mode.
- RecordingActive: Device is currently doing a recording.
- StreamTransferActive: Transfer on the stream is active. e.g. Frame transfer from the device to the host computer is active.
- StageInMotion: Device is currently moving the stage.
- LockInReference: Map selected line to the Lock-In Reference Signal. This is a real-time information coming from the image sensor.

6.8 LineFormat

Name	LineFormat [LineSelector]
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	Unknown
	OpenDrain
	HighSide
	PushPull

Controls the electrical format of the selected physical input or output Line.

- Unknown: unknown line mode.
- OpenDrain: The line is in open drain mode.
- HighSide: The line is in 24V high side switch mode.
- PushPull: the line is in 24V push-pull mode.

6.9 UserOutputSelector

Name	UserOutputSelector
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	
	UserOutput0
	UserOutput1
	UserOutput2
	UserOutput3
	UserOutput4
	UserOutput5
	UserOutput6
	UserOutput7

Select which bit of the User Output register will be set by UserOutputValue.

- UserOutput0: Selects the bit 0 of the User Output register.
- UserOutput1: Selects the bit 1 of the User Output register.
- UserOutput2: Selects the bit 2 of the User Output register.
- UserOutput3: Selects the bit 3 of the User Output register.
- UserOutput4: Selects the bit 4 of the User Output register.
- UserOutput5: Selects the bit 5 of the User Output register.
- UserOutput6: Selects the bit 6 of the User Output register.
- UserOutput7: Selects the bit 7 of the User Output register.

6.10 UserOutputValue

Name	UserOutputValue [UserOutputSelector]
Type	Boolean
Visibility	Expert
AccessMode	RW

Sets the value of the bit selected by UserOutputSelector.

6.11 UserOutputValueAll

Name	UserOutputValueAll
Type	IntReg
Visibility	Expert
AccessMode	RW

Sets the value of all the bits of the User Output register.

ENCODERCONTROL

Category that contains the encoder control features.

7.1 EncoderSelector

Name	EncoderSelector
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	
	Camera
	Stage

Select which encoder to configure.

- Camera: Select Camera (zTag) encoder.
- Stage: Select Stage encoder.

7.2 EncoderInverter

Name	EncoderInverter [EncoderSelector]
Type	Boolean
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW

Invert the count direction of internal encoder counter.

7.3 EncoderReset

Name	EncoderReset [EncoderSelector]
Type	Command
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	WO

Reset the selected encoder counter.

7.4 EncoderValue

Name	EncoderValue [EncoderSelector]
Type	IntReg
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW

Current value of the encoder counter.

7.5 EncoderResolution

Name	EncoderResolution [EncoderSelector]
Type	FloatReg
Visibility	Expert
AccessMode	RO
Unit	nm

Encoder resolution used for selected encoder in nm/tick.

7.6 zTagInverter

Name	zTagInverter
Type	Boolean
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW

Invert the count direction of internal zTag counter. This feature exists for legacy reasons and is an alias of the feature EncoderInverter[Camera].

7.7 zTagReset

Name	zTagReset
Type	Command
Visibility	Beginner
Writable in Acquisition Mode	False
AccessMode	WO

Reset zTag counter.

EVENTCONTROL

Category that contains Event control features.

8.1 EventSelector

Name	EventSelector
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	
	FrameTriggerWait
	FrameTrigger
	FrameTriggerMissed
	FrameStart
	FrameEnd
	FrameTransferStart
	FrameTransferEnd
	RecordingTriggerWait
	RecordingTrigger
	RecordingTriggerMissed
	RecordingStart
	RecordingEnd
	StageMotionStart
	StageMotionEnd

Select which Event to signal to the host application.

- FrameTriggerWait: Device is currently waiting for a FrameStart trigger.
- FrameTrigger: Device just received a trigger to start the capture of one frame.
- FrameTriggerMissed: Device just missed a trigger to start the capture of one frame.
- FrameStart: Device just started the capture of one frame.
- FrameEnd: Device just completed the capture of one frame.
- FrameTransferStart: Device just started the transfer of one frame.
- FrameTransferEnd: Device just completed the transfer of one frame.
- RecordingTriggerWait: Device is currently waiting for a Recording trigger.
- RecordingTrigger: Device just received a recording trigger.
- RecordingTriggerMissed: Device just missed a recording trigger.
- RecordingStart: Device just started the image recording.

- RecordingEnd: Device just completed image recording.
- StageMotionStart: Device just started the stage motion.
- StageMotionEnd: Device just finished the stage motion.

8.2 EventNotification

Name	EventNotification [EventSelector]
Type	Enumeration
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	Off
	On

Activate or deactivate the notification to the host application of the occurrence of the selected Event.

- Off: The selected Event notification is disabled.
- On: The selected Event notification is enabled.

8.3 EventFrameTriggerWaitData

Category that contains all data features related to the FrameTriggerWait Event. All features from this category are event features triggered by the FrameTriggerWait Event. The consumer application could register the events of these features.

8.3.1 EventFrameTriggerWait

Name	EventFrameTriggerWait
Type	IntReg
Visibility	Expert
AccessMode	RO

Returns the unique Identifier of the FrameTriggerWait type of Event. It can be used to register a callback function to be notified of the event occurrence. Its value uniquely identifies the event received.

8.3.2 EventFrameTriggerWaitTimestamp

Name	EventFrameTriggerWaitTimestamp
Type	IntReg
Visibility	Expert
AccessMode	RO
Unit	ns

Returns the Timestamp of the FrameTriggerWait Event. It can be used to determine precisely when the event occurred.

8.4 EventFrameTriggerData

Category that contains all data features related to the FrameTrigger Event. All features from this category are event features triggered by the FrameTrigger Event. The consumer application could register the events of these features.

8.4.1 EventFrameTrigger

Name	EventFrameTrigger
Type	IntReg
Visibility	Expert
AccessMode	RO

Returns the unique Identifier of the FrameTrigger type of Event. It can be used to register a callback function to be notified of the event occurrence. Its value uniquely identifies the event received.

8.4.2 EventFrameTriggerTimestamp

Name	EventFrameTriggerTimestamp
Type	IntReg
Visibility	Expert
AccessMode	RO
Unit	ns

Returns the Timestamp of the FrameTrigger Event. It can be used to determine precisely when the event occurred.

8.5 EventFrameTriggerMissedData

Category that contains all data features related to the FrameTriggerMissed Event. All features from this category are event features triggered by the FrameTriggerMissed Event. The consumer application could register the events of these features.

8.5.1 EventFrameTriggerMissed

Name	EventFrameTriggerMissed
Type	IntReg
Visibility	Expert
AccessMode	RO

Returns the unique Identifier of the FrameTriggerMissed type of Event. It can be used to register a callback function to be notified of the event occurrence. Its value uniquely identifies the event received.

8.5.2 EventFrameTriggerMissedTimestamp

Name	EventFrameTriggerMissedTimestamp
Type	IntReg
Visibility	Expert
AccessMode	RO
Unit	ns

Returns the Timestamp of the FrameTriggerMissed Event. It can be used to determine precisely when the event occurred.

8.6 EventFrameStartData

Category that contains all data features related to the FrameStart Event. All features from this category are event features triggered by the FrameStart Event. The consumer application could register the events of these features.

8.6.1 EventFrameStart

Name	EventFrameStart
Type	IntReg
Visibility	Expert
AccessMode	RO

Returns the unique Identifier of the FrameStart type of Event. It can be used to register a callback function to be notified of the event occurrence. Its value uniquely identifies the event received.

8.6.2 EventFrameStartTimestamp

Name	EventFrameStartTimestamp
Type	IntReg
Visibility	Expert
AccessMode	RO
Unit	ns

Returns the Timestamp of the FrameStart Event. It can be used to determine precisely when the event occurred.

8.7 EventFrameEndData

Category that contains all data features related to the FrameEnd Event. All features from this category are event features triggered by the FrameEnd Event. The consumer application could register the events of these features.

8.7.1 EventFrameEnd

Name	EventFrameEnd
Type	IntReg
Visibility	Expert
AccessMode	RO

Returns the unique Identifier of the FrameEnd type of Event. It can be used to register a callback function to be notified of the event occurrence. Its value uniquely identifies the event received.

8.7.2 EventFrameEndTimestamp

Name	EventFrameEndTimestamp
Type	IntReg
Visibility	Expert
AccessMode	RO
Unit	ns

Returns the Timestamp of the FrameEnd Event. It can be used to determine precisely when the event occurred.

8.8 EventFrameTransferStartData

Category that contains all data features related to the FrameTransferStart Event. All features from this category are event features triggered by the FrameTransferStart Event. The consumer application could register the events of these features.

8.8.1 EventFrameTransferStart

Name	EventFrameTransferStart
Type	IntReg
Visibility	Expert
AccessMode	RO

Returns the unique Identifier of the FrameTransferStart type of Event. It can be used to register a callback function to be notified of the event occurrence. Its value uniquely identifies the event received.

8.8.2 EventFrameTransferStartTimestamp

Name	EventFrameTransferStartTimestamp
Type	IntReg
Visibility	Expert
AccessMode	RO
Unit	ns

Returns the Timestamp of the FrameTransferStart Event. It can be used to determine precisely when the event occurred.

8.9 EventFrameTransferEndData

Category that contains all data features related to the FrameTransferEnd Event. All features from this category are event features triggered by the FrameTransferEnd Event. The consumer application could register the events of these features.

8.9.1 EventFrameTransferEnd

Name	EventFrameTransferEnd
Type	IntReg
Visibility	Expert
AccessMode	RO

Returns the unique Identifier of the FrameTransferEnd type of Event. It can be used to register a callback function to be notified of the event occurrence. Its value uniquely identifies the event received.

8.9.2 EventFrameTransferEndTimestamp

Name	EventFrameTransferEndTimestamp
Type	IntReg
Visibility	Expert
AccessMode	RO
Unit	ns

Returns the Timestamp of the FrameTransferEnd Event. It can be used to determine precisely when the event occurred.

8.10 EventRecordingTriggerWaitData

Category that contains all data features related to the RecordingTriggerWait Event. All features from this category are event features triggered by the RecordingTriggerWait Event. The consumer application could register the events of these features.

8.10.1 EventRecordingTriggerWait

Name	EventRecordingTriggerWait
Type	IntReg
Visibility	Expert
AccessMode	RO

Returns the unique Identifier of the RecordingTriggerWait type of Event. It can be used to register a callback function to be notified of the event occurrence. Its value uniquely identifies the event received.

8.10.2 EventRecordingTriggerWaitTimestamp

Name	EventRecordingTriggerWaitTimestamp
Type	IntReg
Visibility	Expert
AccessMode	RO
Unit	ns

Returns the Timestamp of the RecordingTriggerWait Event. It can be used to determine precisely when the event occurred.

8.11 EventRecordingTriggerData

Category that contains all data features related to the RecordingTrigger Event. All features from this category are event features triggered by the RecordingTrigger Event. The consumer application could register the events of these features.

8.11.1 EventRecordingTrigger

Name	EventRecordingTrigger
Type	IntReg
Visibility	Expert
AccessMode	RO

Returns the unique Identifier of the RecordingTrigger type of Event. It can be used to register a callback function to be notified of the event occurrence. Its value uniquely identifies the event received.

8.11.2 EventRecordingTriggerTimestamp

Name	EventRecordingTriggerTimestamp
Type	IntReg
Visibility	Expert
AccessMode	RO
Unit	ns

Returns the Timestamp of the RecordingTrigger Event. It can be used to determine precisely when the event occurred.

8.12 EventRecordingTriggerMissedData

Category that contains all data features related to the RecordingTriggerMissed Event. All features from this category are event features triggered by the RecordingTriggerMissed Event. The consumer application could register the events of these features.

8.12.1 EventRecordingTriggerMissed

Name	EventRecordingTriggerMissed
Type	IntReg
Visibility	Expert
AccessMode	RO

Returns the unique Identifier of the RecordingTriggerMissed type of Event. It can be used to register a callback function to be notified of the event occurrence. Its value uniquely identifies the event received.

8.12.2 EventRecordingTriggerMissedTimestamp

Name	EventRecordingTriggerMissedTimestamp
Type	IntReg
Visibility	Expert
AccessMode	RO
Unit	ns

Returns the Timestamp of the RecordingTriggerMissed Event. It can be used to determine precisely when the event occurred.

8.13 EventRecordingStartData

Category that contains all data features related to the RecordingStart Event. All features from this category are event features triggered by the RecordingStart Event. The consumer application could register the events of these features.

8.13.1 EventRecordingStart

Name	EventRecordingStart
Type	IntReg
Visibility	Expert
AccessMode	RO

Returns the unique Identifier of the RecordingStart type of Event. It can be used to register a callback function to be notified of the event occurrence. Its value uniquely identifies the event received.

8.13.2 EventRecordingStartTimestamp

Name	EventRecordingStartTimestamp
Type	IntReg
Visibility	Expert
AccessMode	RO
Unit	ns

Returns the Timestamp of the RecordingStart Event. It can be used to determine precisely when the event occurred.

8.14 EventRecordingEndData

Category that contains all data features related to the RecordingEnd Event. All features from this category are event features triggered by the RecordingEnd Event. The consumer application could register the events of these features.

8.14.1 EventRecordingEnd

Name	EventRecordingEnd
Type	IntReg
Visibility	Expert
AccessMode	RO

Returns the unique Identifier of the RecordingEnd type of Event. It can be used to register a callback function to be notified of the event occurrence. Its value uniquely identifies the event received.

8.14.2 EventRecordingEndTimestamp

Name	EventRecordingEndTimestamp
Type	IntReg
Visibility	Expert
AccessMode	RO
Unit	ns

Returns the Timestamp of the RecordingEnd Event. It can be used to determine precisely when the event occurred.

8.15 EventStageMotionStartData

Category that contains all data features related to the StageMotionStart Event. All features from this category are event features triggered by the StageMotionStart Event. The consumer application could register the events of these features.

8.15.1 EventStageMotionStart

Name	EventStageMotionStart
Type	IntReg
Visibility	Expert
AccessMode	RO

Returns the unique Identifier of the StageMotionStart type of Event. It can be used to register a callback function to be notified of the event occurrence. Its value uniquely identifies the event received.

8.15.2 EventStageMotionStartTimestamp

Name	EventStageMotionStartTimestamp
Type	IntReg
Visibility	Expert
AccessMode	RO
Unit	ns

Returns the Timestamp of the StageMotionStart Event. It can be used to determine precisely when the event occurred.

8.16 EventStageMotionEndData

Category that contains all data features related to the StageMotionEnd Event. All features from this category are event features triggered by the StageMotionEnd Event. The consumer application could register the events of these features.

8.16.1 EventStageMotionEnd

Name	EventStageMotionEnd
Type	IntReg
Visibility	Expert
AccessMode	RO

Returns the unique Identifier of the StageMotionEnd type of Event. It can be used to register a callback function to be notified of the event occurrence. Its value uniquely identifies the event received.

8.16.2 EventStageMotionEndTimestamp

Name	EventStageMotionEndTimestamp
Type	IntReg
Visibility	Expert
AccessMode	RO
Unit	ns

Returns the Timestamp of the StageMotionEnd Event. It can be used to determine precisely when the event occurred.

EXTRACTIONCONTROL

Category that contains features for 3D surface extraction configuration. The availability of the features depends on the selected Scan3dExtractionMethod.

9.1 ExtractionMinConfidence

Name	ExtractionMinConfidence
Type	Float
Visibility	Guru
AccessMode	RW
Min	0.0
Max	1.0

Specifies the minimum confidence for the merging/unwrapping of two neighbouring pixels/regions. Higher values produce more conservative and robust, but less precise surfaces.

9.2 ExtractionTimeout

Name	ExtractionTimeout
Type	Float
Visibility	Guru
AccessMode	RW
Min	0.0
Max	10.0

Specifies the maximum time [s] spent for processing algorithm. If a timeout happens, an informational message is placed into the ChunkProcInfoMessage data (if enabled).

9.3 ExtractionExpectedSurfaceFlatness

Name	ExtractionExpectedSurfaceFlatness
Type	Float
Visibility	Guru
AccessMode	RW
Min	0.0
Max	10.0

This setting can be used to tune surface extraction to specific geometries: High values result in higher precision for flat surfaces, at a higher risk of inaccuracy across steps and tilted samples.

9.4 ExtractionConfidenceEpsilon

Name	ExtractionConfidenceEpsilon
Type	Float
Visibility	Guru
AccessMode	RW
Min	0.0
Max	1.0

Higher values increase the robustness (i.e. improve accuracy at the cost of repeatability) in the presence of surfaces with roughness of the order of the wavelength. Values close to 0 are typical.

9.5 ExtractionMismatchEpsilon

Name	ExtractionMismatchEpsilon
Type	Float
Visibility	Guru
AccessMode	RW
Min	0.0
Max	1.0

Higher values increase the robustness (i.e. increase accuracy at the cost of repeatability) in the presence of multi-layer surfaces or shadows. Values close to 0 are typical.

9.6 ExtractionAssumedRMSImageNoise

Name	ExtractionAssumedRMSImageNoise
Type	Float
Visibility	Guru
AccessMode	RW
Min	0.0
Max	100.0

Affects the estimation of the ranging precision in the ImprovedCentreOfMass mode, which is also a core part of the HighPrecisionMode. The value specified here is the RMS noise in each of the I and Q channels individually.

9.7 ExtractionEnvelopeUncertaintyFactor

Name	ExtractionEnvelopeUncertaintyFactor
Type	Float
Visibility	Guru
AccessMode	RW
Min	0.0
Max	5.0

This factor simply scales the estimated uncertainty. It can be used to account for incorrectly modelled noise effects.

9.8 ExtractionPhaseUncertaintyFactor

Name	ExtractionPhaseUncertaintyFactor
Type	Float
Visibility	Guru
AccessMode	RW
Min	0.0
Max	5.0

This factor simply scales the estimated uncertainty. It can be used to account for incorrectly modelled noise effects.

9.9 ExtractionEnvelopeUncertaintyAtLowSignalFactor

Name	ExtractionEnvelopeUncertaintyAtLowSignalFactor
Type	Float
Visibility	Guru
AccessMode	RW
Min	0.0
Max	10.0

This factor scales the estimated uncertainty at low signal strengths, where the amplitude noise starts to be dominated by the constant part rather than scaling proportional with the amplitude.

9.10 ExtractionExpectedEnvelopePhaseMismatch

Name	ExtractionExpectedEnvelopePhaseMismatch
Type	Float
Visibility	Guru
AccessMode	RW
Min	0.0
Max	1.0

Limits the agreement between envelope and phase required to proceed with phase unwrapping. The value is specified as the RMS mismatch between the observed envelope surface and the observed phase surface relative to the demodulation length. By allowing a limited disagreement, the phase unwrapping algorithm may proceed

even if there is a small systematic offset e.g. due to imperfections in the optics, which would be prevented in high-signal areas where the mismatch would indicate model inconsistency otherwise. However, a too large value will prevent unwrapping in lower signal areas, where the measurement uncertainty combined with the expected mismatch leads to ambiguous phase matchings.

9.11 ExtractionPreprocessingFilterType

Name	ExtractionPreprocessingFilterType
Type	Enumeration
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	None
	Median
	Gaussian
	Form

Select the preprocessing surface filtering algorithm applied when using *CenterOfMassPhaseFusion* extraction method. Only available when *Scan3dExtractionMethod* = *CenterOfMassPhaseFusion*

- None: No preprocessing filter is applied.
- Median: A median filter is applied, configured with *ExtractionPreprocessingKernelSize*.
- Gaussian: A gaussian filter is applied, configured with *ExtractionPreprocessingSigma*.
- Form: A form filter is applied, configured with *ExtractionPreprocessingOrder*.

9.12 ExtractionPreprocessingKernelSize

Name	ExtractionPreprocessingKernelSize
Type	Integer
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RW
Min	1
Max	511
Inc	2

Set the kernel footprint size of the (preprocessing) median filter applied on the center of mass surface. The filter is active when *ExtractionPreprocessingFilterType* = *Median* and only available when *Scan3dExtractionMethod* = *CenterOfMassPhaseFusion*.

9.13 ExtractionPreprocessingSigma

Name	ExtractionPreprocessingSigma
Type	Float
Visibility	Guru
AccessMode	RW
Min	0.1
Max	10.0

Set the standard deviation of the (preprocessing) Gaussian kernel filter applied on the center of mass surface. The filter is active when *ExtractionPreprocessingFilterType* = *Gaussian* and only available when *Scan3dExtractionMethod* = *CenterOfMassPhaseFusion*

9.14 ExtractionPreprocessingOrder

Name	ExtractionPreprocessingOrder
Type	Integer
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RW
Min	1
Max	10
Inc	1

Set the order of the polynomial used to fit the center of mass surface. The filter is active when *ExtractionPreprocessingFilterType* = *Form* and only available when *Scan3dExtractionMethod* = *CenterOfMassPhaseFusion*.

9.15 ExtractionCorrectionSigma

Name	ExtractionCorrectionSigma
Type	Float
Visibility	Guru
AccessMode	RW
Min	0.0
Max	100.0

Higher values are beneficial if the sample geometry is flat and has low curvatures.

FPGACONTROL

FPGA and Data Processing configuration features

10.1 ExtSimpMaxHWin

Name	ExtSimpMaxHWin
Type	Integer
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW
Min	1
Max	29
Inc	1

Extended Simple Max half Window Size.

10.2 FPNCorrection

Name	FPNCorrection
Type	Enumeration
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	AverageLastFrames AverageFirstFrames

Defines method applied for sensor fixed pattern noise (FPN) correction. NOTE THAT MEANING AND WORDING ARE SWAPPED (ERRONEOUS).

- AverageLastFrames: Compute average of FIRST frames for pixel-wise offset calculation (e.g. top frames when ScanMode is Down).
- AverageFirstFrames: Compute average of LAST frames for pixel-wise offset calculation (e.g. bottom frames when ScanMode is Up).

10.3 FPNCorrectionNFrames

Name	FPNCorrectionNFrames
Type	Integer
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW
Min	2
Max	64

Number of frames used by averaging FPN correction. Has to be a power of two.

10.4 PeakDetectionMode

Name	PeakDetectionMode
Type	Enumeration
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	GlobalMax
	FirstMax
	LastMax
	FilteredGlobalMax
	FilteredFirstMax
	FilteredLastMax

Select peak detection method used to find amplitude peak value.

- GlobalMax: Find the maximum value across the amplitude a-scan.
- FirstMax: Find first maximum above PeakDetectionThreshold along the amplitude a-scan in direction of recording frames
- LastMax: Find last maximum above PeakDetectionThreshold along the amplitude a-scan in direction of recording frames
- FilteredGlobalMax: Find the maximum value across the filtered amplitude a-scan.
- FilteredFirstMax: Find first maximum above PeakDetectionThreshold along the filtered amplitude a-scan in direction of recording frames
- FilteredLastMax: Find last maximum above PeakDetectionThreshold along the filtered amplitude a-scan in direction of recording frames

10.5 PeakDetectionThreshold

Name	PeakDetectionThreshold
Type	Integer
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW
Min	1
Max	1023
Inc	1

Amplitude threshold used for First/Last peak detection configured by PeakDetectionMode.

10.6 PeakDetectionFilterWidth

Name	PeakDetectionFilterWidth
Type	Integer
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW
Min	1
Max	8
Inc	1

Filter width of moving average applied on amplitude values across an a-scan. The filter is only applied before peak detection. Both a filter width of 1 and a PeakDetectionMode without filter correspond to a disabled filter.

10.7 RecordingActive

Name	RecordingActive
Type	Boolean
Visibility	Expert
AccessMode	RO

The camera is currently recording data.

FIELDBUSCONTROL

Category for the fieldbus communication features.

11.1 FieldbusType

Name	FieldbusType
Type	Enumeration
Visibility	Expert
AccessMode	RO
EnumEntry	
	None
	EtherCAT
	PROFINET

Fieldbus type available on the device.

- None: No fieldbus connection available for this device.
- EtherCAT: EtherCAT connection available on this device.
- PROFINET: PROFINET connection available on this device.

11.2 FieldbusIntDataSelector

Name	FieldbusIntDataSelector
Type	Integer
Visibility	Expert
Min	0
Max	31
Inc	1

Select which Slot of Fieldbus integer data is configured by FieldbusIntDataValue.

11.3 FieldbusIntDataValue

Name	FieldbusIntDataValue [FieldbusIntDataSelector]
Type	IntReg
Visibility	Expert
Writable in Acquisition Mode	True
AccessMode	RW

Access to the Fieldbus integer data. Based on the mode, it is read-only or read/write.

11.4 FieldbusIntDataMode

Name	FieldbusIntDataMode [FieldbusIntDataSelector]
Type	Enumeration
Visibility	Expert
AccessMode	RO
EnumEntry	
	Input
	Output

Check if the selected FieldbusIntData Slot is used as input or output. (camera view)

- Input: The selected slot is used as Input of values.
- Output: The selected slot is used as Output of values.

11.5 FieldbusFloatDataSelector

Name	FieldbusFloatDataSelector
Type	Integer
Visibility	Expert
Min	0
Max	31
Inc	1

Select which Slot of Fieldbus float data is configured by FieldbusFloatDataValue.

11.6 FieldbusFloatDataValue

Name	FieldbusFloatDataValue [FieldbusFloatDataSelector]
Type	FloatReg
Visibility	Expert
Writable in Acquisition Mode	True
AccessMode	RW

Access to the Fieldbus float data. Based on the mode, it is read-only or read/write.

11.7 FieldbusFloatDataMode

Name	FieldbusFloatDataMode [FieldbusFloatDataSelector]
Type	Enumeration
Visibility	Expert
AccessMode	RO
EnumEntry	
	Input
	Output

Check if the selected FieldbusFloatData Slot is used as input or output. (camera view)

- Input: The selected slot is used as Input of values.
- Output: The selected slot is used as Output of values.

11.8 FieldbusBoolSelector

Name	FieldbusBoolSelector
Type	Integer
Visibility	Expert
Min	0
Max	15
Inc	1

Select which boolean value slot of the Fieldbus is configured. (0-7 are camera IN signals and 8-15 are camera OUT signals) See also FieldbusBoolMode description for more information about the event and state information of the FieldbusBool slots.

11.9 FieldbusBoolMode

Name	FieldbusBoolMode [FieldbusBoolSelector]
Type	Enumeration
Visibility	Expert
AccessMode	RO
EnumEntry	
	Input
	Output

Get the information if the selected FieldbusBool Slot is used as input (Fieldbus to Camera) or output (Camera to Fieldbus).

- Input: The selected bool slot is used as camera Input signal. For example: A signal edge could trigger the camera internal acquisition logic (e.g. TriggerSource). Or the signal state could be mapped to control camera functionalities (e.g. LightControllerSource).
- Output: The selected bool slot is used as camera Output signal. It could be used to signalling the current camera state (e.g. see Event And Status Diagram in the appendix of the feature description)

11.10 FieldbusBoolSource

Name	FieldbusBoolSource [FieldbusBoolSelector]
Type	Enumeration
Visibility	Expert
Writable in Acquisition Mode	True
AccessMode	RW
EnumEntry	
	Off
	UserOutput0
	UserOutput1
	UserOutput2
	UserOutput3
	UserOutput4
	UserOutput5
	UserOutput6
	UserOutput7
	FrameTriggerWait
	FrameTriggerMissed
	FrameActive
	RecordingTriggerWait
	RecordingTriggerMissed
	RecordingActive
	StreamTransferActive
	StageInMotion

If the selected FieldbusBool Slot is a camera output signal, this feature is writable and configures which internal state or acquisition signal is mapped to the FieldbusBool slot. The FieldbusState is true if the internal state or signal is active. Otherwise it is false.

- Off: The selected bool slot is disabled. The state is set to false.
- UserOutput0: Map Selected FieldbusBool to UserOutput0.
- UserOutput1: Map Selected FieldbusBool to UserOutput1.
- UserOutput2: Map Selected FieldbusBool to UserOutput2.
- UserOutput3: Map Selected FieldbusBool to UserOutput3.
- UserOutput4: Map Selected FieldbusBool to UserOutput4.
- UserOutput5: Map Selected FieldbusBool to UserOutput5.
- UserOutput6: Map Selected FieldbusBool to UserOutput6.
- UserOutput7: Map Selected FieldbusBool to UserOutput7.
- FrameTriggerWait: Device is currently waiting for a frame start trigger to start a measurement.
- FrameTriggerMissed: Device has missed a FrameStart trigger. Is reset when switching between acquisition and configuration mode.
- FrameActive: Device is currently doing a measurement.
- RecordingTriggerWait: Device is currently waiting for a recording start trigger.
- RecordingTriggerMissed: Device has missed a Recording trigger. Is reset when switching between acquisition and configuration mode.
- RecordingActive: Device is currently doing a recording.
- StreamTransferActive: Transfer on the stream is active. e.g. Frame transfer from the device to the host computer is active.

- StageInMotion: Device is currently moving the stage.

11.11 FieldbusBoolStatus

Name	FieldbusBoolStatus [FieldbusBoolSelector]
Type	Boolean
Visibility	Expert
AccessMode	RO

Returns the current status of the selected FieldbusBool slot.

FILEACCESSCONTROL

Category that contains the File Access control features.

12.1 FileSelector

Name	FileSelector
Type	Enumeration
Visibility	Guru
AccessMode	RW
EnumEntry	
	UserSet0
	UserSet1
	Log

Select the target file in the device. The entries of this enumeration define the names of all files in the device that can be accessed via the File Access.

- UserSet0: The UserSet0 of the device.
- UserSet1: The UserSet1 of the device.
- Log: Read logfiles from the device. This is helpful for debugging device interaction and an advanced feature.

12.2 FileOperationSelector

Name	FileOperationSelector [FileSelector]
Type	Enumeration
Visibility	Guru
AccessMode	RW
EnumEntry	
	Open
	Close
	Read
	Write
	Delete

Select the target operation for the selected file in the device. This Operation is executed when the FileOperationExecute feature is called.

- Open: Opens the file selected by FileSelector in the device. The access mode in which the file is opened is selected by FileOpenMode

- Close: Close the file selected by FileSelector in the device.
- Read: Reads FileAccessLength bytes from the device storage at the file relative offset FileAccessOffset into FileAccessBuffer
- Write: Writes FileAccessLength bytes taken from the FileAccessBuffer into the device storage at the file relative offset FileAccessOffset.
- Delete: Deletes the file selected by FileSelector in the device.

12.3 FileOperationExecute

Name	FileOperationExecute [FileSelector] [FileOperationSelector]
Type	Command
Visibility	Guru
AccessMode	WO

Executes the operation selected by FileOperationSelector on the selected File

12.4 FileOpenMode

Name	FileOpenMode [FileSelector]
Type	Enumeration
Visibility	Guru
AccessMode	RW
EnumEntry	
	Read
	Write
	ReadWrite

Select the access mode in which a file is opened in the device.

- Read: This mode selects read-only open mode
- Write: This mode selects write-only open mode
- ReadWrite: This mode selects read and write open mode

12.5 FileAccessOffset

Name	FileAccessOffset [FileSelector] [FileOperationSelector]
Type	IntReg
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RW
Unit	B

Controls the Offset of the mapping between the device file storage and the FileAccessBuffer.

12.6 FileAccessLength

Name	FileAccessLength [FileSelector] [FileOperationSelector]
Type	IntReg
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RW
Unit	B

Control the Length of the mapping between the device file storage and the FileAccessBuffer.

12.7 FileOperationStatus

Name	FileOperationStatus [FileSelector] [FileOperationSelector]
Type	Enumeration
Visibility	Guru
AccessMode	RO
EnumEntry	Success
	Failure

Represents the file operation status. Upon execution of a successful file operation, it must return Success.

- Success: File Operation was successful.
- Failure: File Operation failed.

12.8 FileOperationResult

Name	FileOperationResult [FileSelector] [FileOperationSelector]
Type	IntReg
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RO

Represents the file operation result. For Read or Write operations, the number of successfully read/written bytes is returned.

12.9 FileSize

Name	FileSize [FileSelector]
Type	IntReg
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RO
Unit	B

Represents the size of the selected file in bytes.

HARDWARECONTROL

Features give hardware information in a low-level style. For example, they are used for compatibility check of UserSets.

13.1 HardwareProcessingBoardType

Name	HardwareProcessingBoardType
Type	IntReg
Visibility	Invisible
AccessMode	RO

Return the value of Processing Board enum (from EEPROM).

13.2 HardwarePIBoardType

Name	HardwarePIBoardType
Type	IntReg
Visibility	Invisible
AccessMode	RO

Return the value of PI Board enum (from EEPROM).

13.3 HardwareStageType

Name	HardwareStageType
Type	IntReg
Visibility	Invisible
AccessMode	RO

Return the value of StageType enum (from EEPROM).

13.4 HardwareIlluminationType

Name	HardwareIlluminationType
Type	IntReg
Visibility	Invisible
AccessMode	RO

Return the value of IlluminationType enum (from EEPROM).

13.5 HardwareSensorType

Name	HardwareSensorType
Type	IntReg
Visibility	Invisible
AccessMode	RO

Return the value of SensorType enum (from EEPROM).

HELIDRIVERCONTROL

Category for heliDriver information and control.

14.1 HeliDriverFamily

Name	HeliDriverFamily
Type	Enumeration
Visibility	Expert
AccessMode	RO
EnumEntry	
	Unknown
	D2
	D3

Get the Family of the connected heliDriver.

- Unknown: heliDriver Family is unknown.
- D2: heliDriver of 2nd generation.
- D3: heliDriver of 3rd generation.

14.2 HeliDriverStatus

Name	HeliDriverStatus
Type	IntReg
Visibility	Guru
AccessMode	RO

Status information of connected heliDriver.

14.3 HeliDriverSerialNumber

Name	HeliDriverSerialNumber
Type	IntReg
Visibility	Guru
AccessMode	RO

Serial Number of connected heliDriver.

14.4 HeliDriverFirmwareVersion

Name	HeliDriverFirmwareVersion
Type	StringReg
Visibility	Guru
AccessMode	RO
Length	16

Version of the firmware of connected heliDriver.

14.5 HeliDriverModuleSelector

Name	HeliDriverModuleSelector
Type	Enumeration
Visibility	Guru
AccessMode	RW
EnumEntry	Module0
	Module1
	Communication

Select the heliDriver Module Slot for reading additional information.

- Module0: heliDriver module in Slot 0
- Module1: heliDriver module in Slot 1
- Communication: heliDriver module in Communication Slot

14.6 HeliDriverModuleType

Name	HeliDriverModuleType [HeliDriverModuleSelector]
Type	Enumeration
Visibility	Guru
AccessMode	RO
EnumEntry	
	RTIO_0
	RTIO_1
	DIO_I
	TRI_LED
	DIO_WLI8HTR
	RTIO_LVDS
	LIA
	Fieldbus
	None

Type of installed heliDriver module.

- RTIO_0: Realtime IO Module for differential encoder signals.
- RTIO_1: Realtime IO Module for single ended encoder signals.
- DIO_I: Digital IO Module including I/O-Link
- TRI_LED: LED Module for controlling three independent current sources.
- DIO_WLI8HTR: Heater module for WLI8.
- RTIO_LVDS: Realtime IO Module for low-voltage differential encoder signals.
- LIA: LIA Module for LockInCam applications.
- Fieldbus: Communication module on heliDriver D3 used for different types of industrial fieldbus communication. Check also feature FieldbusType.
- None: No Module installed.

14.7 HeliDriverModuleFirmwareVersion

Name	HeliDriverModuleFirmwareVersion [HeliDriverModuleSelector]
Type	StringReg
Visibility	Guru
AccessMode	RO
Length	16

Firmware version information of the selected heliDriver module as a hexadecimal string. It returns a string of zeros (00000000), if no firmware version information available for the installed module.

IMAGEFORMATCONTROL

Category for Image Format Control features

15.1 ComponentSelector

Name	ComponentSelector
Type	Enumeration
Visibility	Beginner
AccessMode	RW
EnumEntry	
	Intensity
	Range
	Phase
	Reflectance
	Confidence
	Precision
	UnwrappingGroup
	AdditionalRangeInfo
	FilteredCorrection
	UnwrappedCorrection

The ComponentSelector determines which data acquisition component to control.

- Intensity: The acquisition for surface as integer value is controlled. (Z) Scaling is controlled by Scan3dControl features.
- Range: The acquisition of range (distance) data is controlled. (Z) Data format is controlled by Scan3dControl features.
- Phase: The acquisition of phase information is controlled. (P).
- Reflectance: The acquisition of reflected intensity is controlled. (A).
- Confidence: Configures the output of confidence information, which are currently only available from Clupea, i.e. Scan3dExtractionMode set to ImprovedCentreOfMass or HighPrecisionSurf.
- Precision: Configures the output of precision information given when *Scan3dExtractionMode* set to *ImprovedCentreOfMass* or *HighPrecisionSurf*.
- UnwrappingGroup: Group information of phase algorithm enabled by *Scan3dExtractionMode* set to *HighPrecisionSurf*.
- AdditionalRangeInfo: Return additional range information when *Scan3dExtractionMode* set to *HighPrecisionSurf* or *CenterOfMassPhaseFusion*.

- **FilteredCorrection**: Envelope-phase mismatch with Gaussian filter controlled by **ExtractionCorrectionSigma** applied. Given as angle in $[-\pi, \pi]$. Enabled only when **Scan3dExtractionMode** is set to **CenterOfMassPhaseFusion**.
- **UnwrappedCorrection**: Unwrapped, filtered envelope-phase mismatch, given in um. Enabled only when **Scan3dExtractionMode** is set to **CenterOfMassPhaseFusion**.

15.2 ComponentEnable

Name	ComponentEnable [ComponentSelector]
Type	Boolean
Visibility	Beginner
Writable in Acquisition Mode	False
AccessMode	RW

Controls if the selected component streaming is active. True: Acquisition of component enabled. False: Acquisition of component disabled.

15.3 ComponentIDValue

Name	ComponentIDValue [ComponentSelector]
Type	IntReg
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RO

Return a unique Identifier value that correspond to the selected Component type.

15.4 Width

Name	Width
Type	IntReg
Visibility	Beginner
AccessMode	RO

Width of the image provided by the device (in pixels).

15.5 Height

Name	Height
Type	IntReg
Visibility	Beginner
AccessMode	RO

Height of the image provided by the device (in pixels).

15.6 ReverseX

Name	ReverseX
Type	Boolean
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW

Flip horizontally the image sent by the device.

15.7 ReverseY

Name	ReverseY
Type	Boolean
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW

Flip vertically the image sent by the device.

15.8 Rotation

Name	Rotation
Type	Integer
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW
Min	0
Max	270
Inc	90

Rotate the image sent by the device by 0, 90, 180 or 270 degrees.

15.9 PixelFormat

Name	PixelFormat [ComponentSelector]
Type	Enumeration
Visibility	Beginner
Writable in Acquisition Mode	False
AccessMode	RO
EnumEntry	
	Mono8
	Mono16
	Coord3D_C32f
	Confidence32f

Format of the pixels provided by the device. It represents all the information provided by PixelSize, PixelColorFilter combined in a single feature.

- Mono8: Monochrome 8-bit
- Mono16: Monochrome 16-bit
- Coord3D_C32f: 3D coordinate C 32-bit floating point
- Confidence32f: Confidence 32-bit floating point

15.10 TestPixelMode

Name	TestPixelMode
Type	Enumeration
Visibility	Invisible
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	RelevantPixels
	TestPixels
	FillPixels

Controls the output of additional test pixels and fill patterns.

- RelevantPixels: Only the relevant pixels are given to the user (default).
- TestPixels: In addition to the relevant pixels, the sensor's test pixels are given to the user.
- FillPixels: No pixels are removed and all data are given to the user. This contains the relevant pixels, the sensor test pixels and the fpga fill pattern.

15.11 SensorWidth

Name	SensorWidth
Type	IntReg
Visibility	Expert
AccessMode	RO

Effective width of the sensor in pixels. (considering the Rotation feature)

15.12 SensorHeight

Name	SensorHeight
Type	IntReg
Visibility	Expert
AccessMode	RO

Effective height of the sensor in pixels. (considering the Rotation feature)

15.13 SensorPixelWidth

Name	SensorPixelWidth
Type	FloatReg
Visibility	Guru
AccessMode	RO
Unit	um

Physical size (pitch) in the x direction of a photo sensitive pixel unit.

15.14 SensorPixelHeight

Name	SensorPixelHeight
Type	FloatReg
Visibility	Guru
AccessMode	RO
Unit	um

Physical size (pitch) in the y direction of a photo sensitive pixel unit.

15.15 SensorName

Name	SensorName
Type	StringReg
Visibility	Guru
AccessMode	RO
Length	64

Product name of the imaging Sensor.

LIGHTCONTROL

Category containing the Lighting control features.

16.1 LightControllerSelector

Name	LightControllerSelector
Type	Enumeration
Visibility	Beginner
AccessMode	RW
EnumEntry	
	LightController0
	LED_A
	LED_B
	LED_C

Select the Light Controller to configure.

- LightController0: heliDriver D3 Light Controller for measurement illumination.
- LED_A: heliDriver D3 Light Controller for LED A on the TRI_LED Module.
- LED_B: heliDriver D3 Light Controller for LED B on the TRI_LED Module.
- LED_C: heliDriver D3 Light Controller for LED C on the TRI_LED Module.

16.2 LightControllerSource

Name	LightControllerSource [LightControllerSelector]
Type	Enumeration
Visibility	Beginner
AccessMode	RW
EnumEntry	
	Off
	UserOutput0
	UserOutput1
	UserOutput2
	UserOutput3
	UserOutput4
	UserOutput5
	UserOutput6
	UserOutput7

continues on next page

Table 16.1 – continued from previous page

DIO1
DIO2
DIO3
DIO4
DIO5
DIO6
DIO7
DIO8
DIO10
DIO11
DIO12
DIO13
DIO14
DIO15
DIO16
DIO17
SFIO5
SignalGenerator
FieldbusBool0
FieldbusBool1
FieldbusBool2
FieldbusBool3
FieldbusBool4
FieldbusBool5
FieldbusBool6
FieldbusBool7

Select the input source signal of the Light Controller.

- Off: Disable the Light Controller.
- UserOutput0: Light Controller is controlled by UserOutput0.
- UserOutput1: Light Controller is controlled by UserOutput1.
- UserOutput2: Light Controller is controlled by UserOutput2.
- UserOutput3: Light Controller is controlled by UserOutput3.
- UserOutput4: Light Controller is controlled by UserOutput4.
- UserOutput5: Light Controller is controlled by UserOutput5.
- UserOutput6: Light Controller is controlled by UserOutput6.
- UserOutput7: Light Controller is controlled by UserOutput7.
- DIO1: Light Controller is controlled by DIO1.
- DIO2: Light Controller is controlled by DIO2.
- DIO3: Light Controller is controlled by DIO3.
- DIO4: Light Controller is controlled by DIO4.
- DIO5: Light Controller is controlled by DIO5.
- DIO6: Light Controller is controlled by DIO6.
- DIO7: Light Controller is controlled by DIO7.
- DIO8: Light Controller is controlled by DIO8.
- DIO10: Light Controller is controlled by DIO10.
- DIO11: Light Controller is controlled by DIO11.

- DIO12: Light Controller is controlled by DIO12.
- DIO13: Light Controller is controlled by DIO13.
- DIO14: Light Controller is controlled by DIO14.
- DIO15: Light Controller is controlled by DIO15.
- DIO16: Light Controller is controlled by DIO16.
- DIO17: Light Controller is controlled by DIO17.
- SFIO5: Select the external analog input SFIO5 on the LIA Module as signal source of the Light Controller.
- SignalGenerator: Select the internal SignalGenerator (part of the LIA module) as source of the Light Controller. The Signal Generator is configured by the features where the name starts with SignalGenerator.
- FieldbusBool0: Light Controller is controlled by FieldbusBool0.
- FieldbusBool1: Light Controller is controlled by FieldbusBool1.
- FieldbusBool2: Light Controller is controlled by FieldbusBool2.
- FieldbusBool3: Light Controller is controlled by FieldbusBool3.
- FieldbusBool4: Light Controller is controlled by FieldbusBool4.
- FieldbusBool5: Light Controller is controlled by FieldbusBool5.
- FieldbusBool6: Light Controller is controlled by FieldbusBool6.
- FieldbusBool7: Light Controller is controlled by FieldbusBool7.

16.3 LightCurrentRating

Name	LightCurrentRating [LightControllerSelector]
Type	FloatReg
Visibility	Beginner
Writable in Acquisition Mode	True
AccessMode	RW
Unit	Amp

Configure the maximum current rating of the lighting output. The internal light control (LightController0) has a preconfigured maximum current rating. Is the TriLED module present, the maximum current for LED_A, LED_B and LED_C is configured by this feature. It is required, that the LED is connected to the heliDriver D3 before starting the camera and configure the maximum current.

16.4 LightBrightness

Name	LightBrightness [LightControllerSelector]
Type	Float
Visibility	Beginner
AccessMode	RW
Unit	%
Min	1.0
Max	100.0

Set the brightness in percent of the illumination output.

16.5 LightCurrentMeasured

Name	LightCurrentMeasured [LightControllerSelector]
Type	FloatReg
Visibility	Beginner
AccessMode	RO
Unit	Amp

The measured current applied to the lighting in Amp.

16.6 SignalGeneratorFrequency

Name	SignalGeneratorFrequency
Type	Float
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RW
Unit	Hz
Min	100.0
Max	50000.0

Configure the signal generator frequency if LightControllerSource=SignalGenerator.

16.7 SignalGeneratorAmplitude

Name	SignalGeneratorAmplitude
Type	Float
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RW
Unit	%
Min	1.0
Max	100.0

Configure the amplitude of the internal generated signal if LightControllerSource=SignalGenerator.

16.8 SignalGeneratorOffset

Name	SignalGeneratorOffset
Type	Float
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RW
Unit	%
Min	1.0
Max	100.0

Configure the offset of the internal generated signal if LightControllerSource=SignalGenerator.

16.9 SignalGeneratorModulationMode

Name	SignalGeneratorModulationMode
Type	Enumeration
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	Off
	On
	Pulsed

Configure the operation mode for the modulation of the signal generator if LightControllerSource=SignalGenerator.

- Off: The modulation of the signal generator output is disabled and only the DC component is on the current output.
- On: The modulation of the signal generator output is always enabled.
- Pulsed: The modulation of signal generator output is triggered by an external signal and enabled for the time configured in SignalGeneratorPulseDuration. Otherwise only the DC current is on the output.

16.10 SignalGeneratorMode

Name	SignalGeneratorMode
Type	Enumeration
Visibility	Invisible
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	Off
	On
	Pulsed

This feature is Deprecated!

Configure the operation mode for the modulation of the signal generator if LightControllerSource=SignalGenerator.

- Off: The modulation of the signal generator output is disabled and only the DC component is on the current output.
- On: The modulation of the signal generator output is always enabled.
- Pulsed: The modulation of signal generator output is triggered by an external signal and enabled for the time configured in SignalGeneratorPulseDuration. Otherwise only the DC current is on the output.

16.11 SignalGeneratorPulseDuration

Name	SignalGeneratorPulseDuration
Type	Float
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RW
Unit	us
Min	4.0
Max	1000000.0

Configure the duration of the generated signal if the SignalGeneratorModulationMode is Pulsed and LightControllerSource=SignalGenerator.

16.12 SignalGeneratorTriggerSource

Name	SignalGeneratorTriggerSource
Type	Enumeration
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	Off
	RTIO2
	RTIO3
	RTIO6
	RTIO7
	FI2
	FI3

Specifies the internal signal or physical input line to use as the signal generator trigger source, if LightControllerSource=SignalGenerator and SignalGeneratorModulationMode=Pulsed.

- Off: The SignalGenerator has no external trigger source configured.
- RTIO2: Specifies that the trigger source for the SignalGenerator pulse is generated by RTIO2.
- RTIO3: Specifies that the trigger source for the SignalGenerator pulse is generated by RTIO3.
- RTIO6: Specifies that the trigger source for the SignalGenerator pulse is generated by RTIO6.
- RTIO7: Specifies that the trigger source for the SignalGenerator pulse is generated by RTIO7.
- FI2: Specifies that the trigger source for the SignalGenerator pulse is generated by FI2.
- FI3: Specifies that the trigger source for the SignalGenerator pulse is generated by FI3.

16.13 SignalGeneratorTriggerActivation

Name	SignalGeneratorTriggerActivation
Type	Enumeration
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	RisingEdge
	FallingEdge
	AnyEdge

Specifies the activation mode of the trigger if LightControllerSource=SignalGenerator and SignalGeneratorModulationMode=Pulsed.

- RisingEdge: Specifies that the trigger is considered valid on the rising edge of the source signal.
- FallingEdge: Specifies that the trigger is considered valid on the falling edge of the source signal.
- AnyEdge: Specifies that the trigger is considered valid on the rising or falling edge of the source signal.

16.14 SignalGeneratorTriggerDelay

Name	SignalGeneratorTriggerDelay
Type	Float
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RW
Unit	us
Min	0.0625
Max	10000000.0

Configure the delay appended to a trigger signal before the signal generator pulse is started, if LightControllerSource=SignalGenerator and SignalGeneratorModulationMode=Pulsed.

16.15 SignalGeneratorPhaseShift

Name	SignalGeneratorPhaseShift
Type	Float
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RW
Unit	°
Min	0.0
Max	360.0
Inc	0.087890625

Configure the start phase in degrees of the generated signal in pulsed mode. It is only relevant for LightControllerSource=SignalGenerator and SignalGeneratorModulationMode=Pulsed.

LOCKINCONTROL

Category that contains the LockIn Camera control features.

17.1 LockInTargetReferenceFrequency

Name	LockInTargetReferenceFrequency
Type	Float
Visibility	Expert
Operation Mode	LockInCam
AccessMode	RW
Unit	Hz
Min	306.0
Max	depends on feature <i>LockInReferenceFrequencyMax</i>

Specify the internally generated reference frequency, if LockInReferenceSourceType is Internal or Reference trigger is inactive. For External LockInReferenceSourceType or enabled Reference trigger, this value is used for future calculations as expected reference frequency.

17.2 LockInActualReferenceFrequency

Name	LockInActualReferenceFrequency
Type	FloatReg
Visibility	Expert
Operation Mode	LockInCam
AccessMode	RO
Unit	Hz

If LockInReferenceSourceType is set to Internal or Reference trigger is inactive, this feature returns the exact configured reference frequency (demodulation frequency). This value may differ slightly from the configured LockInTargetReferenceFrequency.

17.3 LockInExpectedFrequencyDeviation

Name	LockInExpectedFrequencyDeviation
Type	Integer
Visibility	Expert
Operation Mode	LockInCam
AccessMode	RW
Unit	%
Min	0
Max	100

Specifies the tolerance (+/-) in percent of the expected external reference frequency, if LockInReferenceSourceType is set to External or Reference trigger is enabled.

17.4 LockInTargetTimeConstantNPeriods

Name	LockInTargetTimeConstantNPeriods
Type	Integer
Visibility	Expert
Operation Mode	LockInCam
AccessMode	RW
Min	1
Max	100

If DeviceConfigurationMode=LockInCam, this feature configures the filter time constant of the lock-in process, i.e., the number of periods per frame during which demodulation is active. The total number of periods per frame, which is roughly the reciprocal of the frame rate of the camera, is the sum of LockInActualTimeConstantNPeriods + LockInActualBlankDurationNPeriods. A period is the inverse of LockInActualReferenceFrequency.

17.5 LockInActualTimeConstantNPeriods

Name	LockInActualTimeConstantNPeriods
Type	IntReg
Visibility	Expert
Operation Mode	LockInCam
AccessMode	RO

If DeviceOperationMode=LockInCam, this feature returns the effective lock-in filter time constant, i.e., the number of demodulation periods per frame. The target value is configured using LockInTargetTimeConstantNPeriods.

17.6 LockInTargetBlankDurationNPeriods

Name	LockInTargetBlankDurationNPeriods
Type	Integer
Visibility	Expert
Operation Mode	LockInCam
AccessMode	RW
Min	0
Max	100

If DeviceConfigurationMode=LockInCam, this feature configures the number of blank periods, i.e., the number of periods per frame during which demodulation is inactive. The total number of periods per frame, which is roughly the reciprocal of the camera frame rate, is the sum of LockInActualTimeConstantNPeriods + LockInActualBlankDurationNPeriods. LockInTargetBlankDurationNPeriods can thus be used to decrease the frame rate without changing the lock-in filter time constant. One cycle lasts one period at LockInActualReferenceFrequency.

17.7 LockInActualBlankDurationNPeriods

Name	LockInActualBlankDurationNPeriods
Type	IntReg
Visibility	Expert
Operation Mode	LockInCam
AccessMode	RO

If DeviceOperationMode=LockInCam, this feature returns the effective number of blank periods per frame when DeviceOperationMode=LockInCam. The target value is configured using LockInTargetBlankDurationNPeriods.

17.8 LockInReferenceSourceType

Name	LockInReferenceSourceType
Type	Enumeration
Visibility	Expert
Operation Mode	LockInCam
AccessMode	RW
EnumEntry	Internal
	External

Specifies the reference signal source when DeviceOperationMode=LockInCam. It is an alias of the feature TriggerMode[Reference].

- Internal: Specifies the internal reference source.
- External: Specifies the external reference source. The external source input is configured by LockInReferenceSourceSignal.

17.9 LockInReferenceSourceSignal

Name	LockInReferenceSourceSignal
Type	Enumeration
Visibility	Expert
Operation Mode	LockInCam
AccessMode	RW
EnumEntry	
	Other
	FI2
	FI3
	RTIO6
	RTIO7

Specifies the input line of the external reference signal if its source is set to External and the DeviceOperationMode=LockInCam is active. It is an alias of the feature TriggerSource[Reference].

- Other: Placeholder for another configuration which does not match the alias to TriggerSource[Reference].
- FI2: Specifies that the reference signal is from provided to input FI2. (recommended)
- FI3: Specifies that the reference signal is from provided to input FI3.
- RTIO6: Specifies that the reference signal is provided to input RTIO6. (Only for reference frequencies less than 10kHz)
- RTIO7: Specifies that the reference signal is provided to input RTIO7. (Only for reference frequencies less than 10kHz)

17.10 LockInReferenceTimeShift

Name	LockInReferenceTimeShift
Type	Float
Visibility	Expert
Operation Mode	LockInCam
AccessMode	RW
Unit	us
Min	0.0
Max	489.6
Inc	0.015

Adds a time delay to the external reference signal with a resolution of 5ns if its source is set to External and the DeviceOperationMode=LockInCam is active. It is an alias of the feature TriggerDelay[Reference].

17.11 LockInReferenceFrequencyScaler

Name	LockInReferenceFrequencyScaler
Type	Enumeration
Visibility	Expert
Operation Mode	LockInCam
AccessMode	RW
EnumEntry	
	Off
	MultiplyBy4
	MultiplyBy2
	DivideBy2
	DivideBy4
	DivideBy8
	DivideBy16
	DivideBy32
	DivideBy64

Specifies a frequency scaler factor for an external reference signal if its source is set to External and the DeviceOperationMode=LockInCam is active. It is an alias of the features TriggerDivider[Reference] and TriggerMultiplier[Reference]. The device performs frequency synthesis based on these features. When the value is DivideBy4 frequency synthesis is disabled and every reference trigger (falling and/or rising edges depending on TriggerActivation[Reference]) triggers a quarter period of (sampled data) demodulation.

- Off: The reference frequency is not adjusted by the camera.
- MultiplyBy4: The reference frequency is multiplied by 4.
- MultiplyBy2: The reference frequency is multiplied by 2.
- DivideBy2: The reference frequency is divided by 2.
- DivideBy4: The reference frequency is divided by 4.
- DivideBy8: The reference frequency is divided by 8.
- DivideBy16: The reference frequency is divided by 16.
- DivideBy32: The reference frequency is divided by 32.
- DivideBy64: The reference frequency is divided by 64.

17.12 LockInCoupling

Name	LockInCoupling
Type	Enumeration
Visibility	Expert
Operation Mode	LockInCam
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	DC
	AC

Specify the signal coupling method when DeviceOperationMode=LockInCam. It is an alias of the feature BackgroundSuppression.

- DC: Turning on DC coupling.

- AC: Turning on AC coupling.

17.13 LockInSensitivity

Name	LockInSensitivity
Type	Float
Visibility	Expert
Operation Mode	LockInCam
Writable in Acquisition Mode	False
AccessMode	RW
Min	0.0
Max	1.0

Specifies the sensitivity range of 0.0 - 1.0. While 1.0 means the highest sensitivity and results in maximum signal amplitude. This is an alias of the feature ExposureRatio.

MISCCONTROL

Category that contains miscellaneous features.

18.1 LogSelector

Name	LogSelector
Type	Enumeration
Visibility	Guru
AccessMode	RW
EnumEntry	
	FeatureAccess
	System
	Statemachines
	HardwareAccess

Select the internal logging group for additional configuration.

- FeatureAccess: Configure the logging for feature access.
- System: Configure the loggers for general system information.
- Statemachines: Configure the loggers of the internal state machines.
- HardwareAccess: Configure the loggers for hardware access.

18.2 LogLevel

Name	LogLevel [LogSelector]
Type	Enumeration
Visibility	Guru
AccessMode	RW
EnumEntry	
	critical
	error
	warning
	notice
	info
	debug

Specifies the minimum detail level of internal logging information.

- critical: log critical errors.

- error: log non-critical errors.
- warning: log warnings.
- notice: log normal, but significant events.
- info: log information messages.
- debug: log debug messages.

18.3 LastErrorMessage

Name	LastErrorMessage
Type	StringReg
Visibility	Invisible
AccessMode	RO
Length	512

Last internal error message from the heliService.

18.4 DeviceOperationMode

Name	DeviceOperationMode
Type	Enumeration
Visibility	Beginner
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	Topography
	Tomography
	LockInCam

Select operation mode of the heliotis device.

- Topography: The device operates as 3D Measurement system and returns topography information to the user. The device configuration is based on motion information (e.g. position, speed, distance) and done with features from the WLIScanner category.
- Tomography: The device operates in Tomography mode and returns volume information to the user. The device configuration is based on motion information (e.g. position, speed, distance) and done with features from the WLIScanner category.
- LockInCam: The device operates in LockInCam mode and returns a burst of demodulating frames to the user. The device configuration is based on frequency information and done by the LockIn features.

18.5 LowLevelMode

Name	LowLevelMode
Type	Enumeration
Visibility	Guru
EnumEntry	
	Off
	On

Configure the low-level feature access mode.

- Off: Low-Level feature access is disabled.
- On: A basic low-level mode is active. This allows to write specific low-level features. Feature value dependent calculations are still done according to the DeviceOperationMode setting.

MOTIONCONTROL

Motion configuration features

19.1 StageType

Name	StageType
Type	Enumeration
Visibility	Beginner
AccessMode	RO
EnumEntry	
	Unknown
	Disabled
	UltraEncoderType2
	StandardEncoderType1
	UltraXenax
	StandardXenax
	StandardEncoderType4
	StandardEncoderType3
	Custom

Get the installed Stage Type used and controlled by heliService

- Unknown: Motion Type is unknown / not configured.
- Disabled: Motion is not available on this device.
- UltraEncoderType2: Motion with high precision encoder is available.
- StandardEncoderType1: Motion with standard encoder is available.
- UltraXenax: Motion controlled by external Xenax Controller with 100nm high precision encoder.
- StandardXenax: Motion controlled by external Xenax Controller with 1um encoder.
- StandardEncoderType4: Motion with standard encoder is available.
- StandardEncoderType3: Motion with standard encoder is available.
- Custom: Custom motion control is used.

19.2 zTagResolution

Name	zTagResolution
Type	FloatReg
Visibility	Expert
AccessMode	RO

zTag Resolution in nm/tick

19.3 ScanMode

Name	ScanMode
Type	Enumeration
Visibility	Beginner
AccessMode	RW
EnumEntry	
	None
	DownAndWait
	UpAndWait
	Smart
	Down
	Up

Select scanning mode used for data acquisition. In a measurement/acquisition sequence the selected mode is used for controlling the motion.

- None: Motion is disabled.
- DownAndWait: Measure the surface when move downward. Wait at the end position after the measurement.
- UpAndWait: Measure the surface when move upward. Wait at the end position after the measurement.
- Smart: Select the closest position as start position and measure up or downward. Wait at the end position after the measurement. This Mode is not supported when multiple segments are active (VerticalSegmentEnable).
- Down: Measure the surface when move downward. Move back to the start position after the measurement.
- Up: Measure the surface when move upward. Move back to the start position after the measurement.

19.4 Position

Name	Position
Type	Float
Visibility	Beginner
AccessMode	RW
Inc	0.001
Min	depends on feature <i>PositionMin</i>
Max	depends on feature <i>PositionMax</i>

Current position [mm] of the stage. Write to this feature would set the TargetPosition and move there as non blocking execution. The motion is executed with the speed configured in GeneralSpeed feature. Reading this feature returns the current position of the stage.

19.5 TargetPosition

Name	TargetPosition
Type	Float
Visibility	Beginner
AccessMode	RW
Inc	0.001
Min	depends on feature <i>PositionMin</i>
Max	depends on feature <i>PositionMax</i>

Target position [mm] of the stage. Write to this feature would move to that position as non blocking execution.

19.6 MotionMode

Name	MotionMode
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	Off
	Immediate

Configure the behavior what happens when a new TargetPosition/Position is written.

- Off: Writing the TargetPosition/Position features has no effect to the current stage motion.
- Immediate: The stage moves to the new TargetPosition directly after feature write.

19.7 StartPosition

Name	StartPosition
Type	FloatReg
Visibility	Expert
AccessMode	RO

Next measurement start position [mm], if nothing changes until next measurement.

19.8 EndPosition

Name	EndPosition
Type	FloatReg
Visibility	Expert
AccessMode	RO

Next measurement end position [mm], if nothing changes until next measurement.

19.9 TriggerPosition

Name	TriggerPosition
Type	FloatReg
Visibility	Expert
AccessMode	RO

Next measurement trigger position [mm], if nothing changes until next measurement.

19.10 GeneralSpeed

Name	GeneralSpeed
Type	Float
Visibility	Beginner
AccessMode	RW
Min	0.1
Max	depends on feature <i>SpeedMax</i>

Configure the general motion speed [mm/s].

19.11 AccelerationRamp

Name	AccelerationRamp
Type	Float
Visibility	Expert
AccessMode	RW
Min	0.02
Max	depends on feature <i>AccelerationRampMax</i>

Acceleration ramp used before and after the scan [mm]. This distance is used to accelerate and decelerate the stage before and after a scan.

19.12 AutomatedStageInit

Name	AutomatedStageInit
Type	Boolean
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW

Controls if the stage is initialized automated when the camera is booted or reset. This is useful in combination with UserSets.

19.13 StageInit

Name	StageInit
Type	Command
Visibility	Beginner
Writable in Acquisition Mode	False
AccessMode	WO

Initialize the motion.

19.14 StageMove

Name	StageMove
Type	Command
Visibility	Guru
AccessMode	WO

Move the stage immediately to the written TargetPosition/Position.

19.15 StageCategory

Name	StageCategory
Type	Enumeration
Visibility	Beginner
AccessMode	RO
EnumEntry	Unknown
	PositionBased

Get the stage category of the current used stage

- Unknown: Stage category not known.
- PositionBased: Position based stage. heliService controls the stage over target positions.

19.16 MoLibName

Name	MoLibName
Type	StringReg
Visibility	Expert
AccessMode	RO
Length	16

Name of used motion library.

19.17 MoLibVersion

Name	MoLibVersion
Type	StringReg
Visibility	Expert
AccessMode	RO
Length	16

Version of used motion library.

19.18 SoftlimitSelector

Name	SoftlimitSelector
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	
	HardwareLower
	HardwareUpper
	UserLower
	UserUpper

Select the stage softlimit to configure.

- HardwareLower: Lower hardware limit of the stage.
- HardwareUpper: Upper hardware limit of the stage.
- UserLower: Lower softlimit of the stage given by the user.
- UserUpper: Upper softlimit of the stage given by the user.

19.19 SoftlimitEnable

Name	SoftlimitEnable [SoftlimitSelector]
Type	Boolean
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW

Controls if the selected softlimit is active. True: Selected softlimit is checked. False: Selected softlimit is ignored.

19.20 SoftlimitPosition

Name	SoftlimitPosition [SoftlimitSelector]
Type	Float
Visibility	Expert
Writable in Acquisition Mode	True
AccessMode	RW
Inc	0.1
Min	depends on feature <i>SoftlimitMin</i>
Max	depends on feature <i>SoftlimitMax</i>

Configure the selected softlimit position [mm].

19.21 SoftlimitSave

Name	SoftlimitSave [SoftlimitSelector]
Type	Command
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	WO

Store the selected softlimit in non-volatile camera memory.

19.22 StageInitialized

Name	StageInitialized
Type	Boolean
Visibility	Expert

Indicates whether the stage is initialized and ready to move.

19.23 StageInMotion

Name	StageInMotion
Type	Boolean
Visibility	Expert

Indicates whether the stage is in motion.

19.24 StageStatusSelector

Name	StageStatusSelector
Type	Enumeration
Visibility	Guru
AccessMode	RW
EnumEntry	
	Initialized
	InMotion

Select the motion module status to read using StageStatus feature.

- Initialized: Stage is initialized and ready to move.
- InMotion: Indicate if the stage is currently moving.

19.25 StageStatus

Name	StageStatus [StageStatusSelector]
Type	Boolean
Visibility	Guru
AccessMode	RO

Reads the status of the motion module selected by StageStatusSelector.

MOTIONFEATURES

Stage specific motion features (optional).

20.1 MoDummy

This subchapter contains the features available from the motion library (MoLibName) MoDummy in version (MoLibVersion) 1.6.0.

20.1.1 UserZTagResolution

Name	UserZTagResolution
Type	FloatReg
Visibility	Beginner
AccessMode	RW

Configure the zTag resolution of the libMoDummy library. This is used when the zTag information is used for surface scaling.

20.2 MoH8

This subchapter contains the features available from the motion library (MoLibName) MoH8 in version (MoLibVersion) 1.6.0.

20.2.1 StageInitMode

Name	StageInitMode
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	Default
	Force
	Smart

Configure initialisation mode of H8 stage.

- Default: Use default initialisation method.

- Force: Force a reinitialisation of the stage.
- Smart: Smart stage initialisation, don't initialize the stage if already done.

20.2.2 MC5004ProxyIndex

Name	MC5004ProxyIndex
Type	IntReg
Visibility	Expert
AccessMode	RW

Index of low-level controller access.

20.2.3 MC5004ProxySubindex

Name	MC5004ProxySubindex
Type	IntReg
Visibility	Expert
AccessMode	RW

Subindex of low-level controller access.

20.2.4 MC5004ProxySize

Name	MC5004ProxySize
Type	IntReg
Visibility	Expert
AccessMode	RW

Size of low-level controller access.

20.2.5 MC5004ProxyValue

Name	MC5004ProxyValue
Type	IntReg
Visibility	Expert
AccessMode	RW

Value of low-level controller access.

20.2.6 MC5004Read

Name	MC5004Read
Type	Command
Visibility	Expert
AccessMode	RO

Read command for MC5004 low-level access.

20.2.7 MC5004Write

Name	MC5004Write
Type	Command
Visibility	Expert
AccessMode	RO

Write command for MC5004 low-level access.

20.2.8 iCtw8ProxyAddress

Name	iCtw8ProxyAddress
Type	IntReg
Visibility	Expert
AccessMode	RW

Address of low-level interpolator access.

20.2.9 iCtw8ProxySize

Name	iCtw8ProxySize
Type	IntReg
Visibility	Expert
AccessMode	RW

Size of low-level interpolator access.

20.2.10 iCtw8ProxyValue

Name	iCtw8ProxyValue
Type	IntReg
Visibility	Expert
AccessMode	RW

Value of low-level interpolator access.

20.2.11 iCtw8Read

Name	iCtw8Read
Type	Command
Visibility	Expert
AccessMode	RO

Read command for iCtw8 low-level access.

20.2.12 iCtw8Write

Name	iCtw8Write
Type	Command
Visibility	Expert
AccessMode	RO

Write command for iCtw8 low-level access.

20.3 MoXenax

This subchapter contains the features available from the motion library (MoLibName) MoXenax in version (MoLib-Version) 1.6.0.

20.3.1 StageInitMode

Name	StageInitMode
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	
	Default
	Force
	Smart

Configure initialisation mode of H8 stage.

- Default: Use default initialisation method.
- Force: Force a reinitialisation of the stage.
- Smart: Smart stage initialisation, don't initialize the stage if already done.

20.3.2 XenaxIP

Name	XenaxIP
Type	StringReg
Visibility	Beginner
AccessMode	RW
Length	16

Configure the IP Address of the Xenax controller used for z scan.

20.3.3 XenaxPort

Name	XenaxPort
Type	IntReg
Visibility	Beginner
AccessMode	RW

Configure the TCP Port of the Xenax controller used for z scan.

20.3.4 XenaxCommand

Name	XenaxCommand
Type	StringReg
Visibility	Guru
AccessMode	RW
Length	16

Low-Level Command sent to Xenax controller in next XenaxSubmit command.

20.3.5 XenaxReply

Name	XenaxReply
Type	StringReg
Visibility	Guru
AccessMode	RO
Length	16

Reply of last Low-Level Command sent to Xenax controller.

20.3.6 XenaxSubmit

Name	XenaxSubmit
Type	Command
Visibility	Guru
AccessMode	RO

Execute command in XenaxCommand on Xenax controller.

OPTICCONTROL

Category for optic information and control.

21.1 OpticControllerSelector

Name	OpticControllerSelector
Type	Enumeration
Visibility	Beginner
AccessMode	RW
EnumEntry	
	Fixed
	TemperatureController

Controls if the optic temperature control is active.

- Fixed: Configure the fixed part of the optics.
- TemperatureController: Configure the temperature controller of the optics.

21.2 OpticType

Name	OpticType [OpticControllerSelector]
Type	Enumeration
Visibility	Beginner
AccessMode	RW
EnumEntry	
	Custom
	WLI8_X2
	WLI8_X4
	WLI8_X8
	MIRAU_X10
	MIRAU_X20
	MIRAU_X50
	MIRAU_X100
	LM_MIRAU_X10
	LM_MIRAU_X20
	LM_MIRAU_X50
	MT_MIRAU_X20
	MT_MIRAU_X50
	H9L_X0_5
	H9L_X0_8
	H9L_X1_0
	H9L_X2_0
	H9S_X0_5
	H9S_X0_8
	H9S_X1_0
	H9S_X1_5
	H9S_X2_0
	H9S_X3_0

Select the optic type to load predefined configurations. This writes implicit the features Magnification, NumericalAperture, WavelengthCorrection and TargetVerticalSpacing. If a user application changes one of this optics relevant features, the OpticType changes to Custom. This feature is only available for OpticControlSelector=Fixed.

- Custom: The optic relevant parameters are configured by the user.
- WLI8_X2: Select the heliInspect H8 WLI8-X2 optic.
- WLI8_X4: Select the heliInspect H8 WLI8-X4 optic.
- WLI8_X8: Select the heliInspect H8 WLI8-X8 optic.
- MIRAU_X10: **(Deprecated!)** Select the heliInspect H8 MIRAU-X10 optic.
- MIRAU_X20: **(Deprecated!)** Select the heliInspect H8 MIRAU-X20 optic.
- MIRAU_X50: **(Deprecated!)** Select the heliInspect H8 MIRAU-X50 optic.
- MIRAU_X100: **(Deprecated!)** Select the heliInspect H8 MIRAU-X100 optic.
- LM_MIRAU_X10: Select the heliInspect H8 WLI8-Mirau + Leitz 10x optic.
- LM_MIRAU_X20: Select the heliInspect H8 WLI8-Mirau + Leitz 20x optic.
- LM_MIRAU_X50: Select the heliInspect H8 WLI8-Mirau + Leitz 50x optic.
- MT_MIRAU_X20: Select the heliInspect H8 WLI8-Mirau + Mitutoyo WLI Plan Apo 10x optic.
- MT_MIRAU_X50: Select the heliInspect H8 WLI8-Mirau + Mitutoyo WLI Plan Apo 25x optic.
- H9L_X0_5: Select the heliInspect H9 H9L-X0.5 optic type.

- H9L_X0_8: Select the heliInspect H9 H9L-X0.8 optic type.
- H9L_X1_0: Select the heliInspect H9 H9L-X1.0 optic type.
- H9L_X2_0: Select the heliInspect H9 H9L-X2.0 optic type.
- H9S_X0_5: Select the heliInspect H9 H9S-X0.5 optic type.
- H9S_X0_8: Select the heliInspect H9 H9S-X0.8 optic type.
- H9S_X1_0: Select the heliInspect H9 H9S-X1.0 optic type.
- H9S_X1_5: Select the heliInspect H9 H9S-X1.5 optic type.
- H9S_X2_0: Select the heliInspect H9 H9S-X2.0 optic type.
- H9S_X3_0: Select the heliInspect H9 H9S-X3.0 optic type.

21.3 OpticControllerSerialNumber

Name	OpticControllerSerialNumber [OpticControllerSelector]
Type	StringReg
Visibility	Beginner
AccessMode	RO
Length	16

Serial Number of the optic controller.

21.4 OpticControllerFirmwareVersion

Name	OpticControllerFirmwareVersion [OpticControllerSelector]
Type	StringReg
Visibility	Beginner
AccessMode	RO
Length	16

Firmware Version of the optic controller.

21.5 NumericalAperture

Name	NumericalAperture [OpticControllerSelector]
Type	FloatReg
Visibility	Beginner
AccessMode	RW

NumericalAperture of the optic.

21.6 Magnification

Name	Magnification [OpticControllerSelector]
Type	FloatReg
Visibility	Beginner
AccessMode	RW

Magnification of the optic.

21.7 WorkingDistance

Name	WorkingDistance
Type	FloatReg
Visibility	Beginner
AccessMode	RO

If the OpticType has been configured, the corresponding working distance in mm can be read back.

21.8 OpticTemperatureStatus

Name	OpticTemperatureStatus [OpticControllerSelector]
Type	Enumeration
Visibility	Expert
AccessMode	RO
EnumEntry	
	NotConnected
	Ready
	Active

Reads the status of the temperature controller.

- NotConnected: The temperature controller is physically not connected.
- Ready: The temperature controller is ready to use.
- Active: The temperature controller is active.

21.9 OpticTemperatureControlMode

Name	OpticTemperatureControlMode [OpticControllerSelector]
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	
	Off
	On

Controls if the optic temperature control is active.

- Off: Disables the optic heater.
- On: Enable the optic heater.

21.10 OpticTargetTemperature

Name	OpticTargetTemperature [OpticControllerSelector]
Type	FloatReg
Visibility	Expert
AccessMode	RW
Unit	C

Define the target temperature for the optic.

LowLevel Sensor Registers

SCAN3DCONTROL

Category for control of 3D camera specific features.

23.1 Scan3dExtractionMethod

Name	Scan3dExtractionMethod
Type	Enumeration
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	Default
	rawIQ
	rawAmp
	rawExtSimpMax
	centerOfMass
	centerOfMassIQCorrection
	AcceleratedCenterOfMass
	AcceleratedCenterOfMassIQCorrection
	ImprovedCenterOfMass
	HighPrecision
	CenterOfMassPhaseFusion

Select the method for extracting 3D surface data from the raw sensor data. This feature configures both the FPGA and the diaphus processing.

- Default: Configure the default method, recommended by heliotis, for surface extraction.
- rawIQ: Disable processing and submit sensor raw data.
- rawAmp: FPGA based amplitude volume calculation.
- rawExtSimpMax: FPGA based calculation of extended simple max.
- centerOfMass: CPU based center of mass surface extraction.
- centerOfMassIQCorrection: CPU based center of mass surface extraction with additional IQ correction.
- AcceleratedCenterOfMass: FPGA based center of mass surface extraction.
- AcceleratedCenterOfMassIQCorrection: FPGA based center of mass surface extraction with additional IQ correction.
- ImprovedCenterOfMass: CPU based improved center of mass surface extraction.
- HighPrecision: CPU based high precision surface extraction.

- CenterOfMassPhaseFusion: CPU based surface extraction combining classical center of mass with phase data evaluation.

23.2 Scan3dScalingMethod

Name	Scan3dScalingMethod
Type	Enumeration
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	FrameThickness
	zTags

Specifies the method used to scale the raw surface to distance units.

- FrameThickness: Scale surface based on frame thickness.
- zTags: Scale surface based on recorded position information.

23.3 Scan3dDistanceUnit

Name	Scan3dDistanceUnit
Type	Enumeration
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	Frames
	nm
	um
	mm

Specifies the unit used when delivering calibrated distance data.

- Frames: Frame based surface value.
- nm: Distance values are in nanometer units.
- um: Distance values are in micrometer units.
- mm: Distance values are in millimeter units.

23.4 Scan3dCoordinateSelector

Name	Scan3dCoordinateSelector
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	CoordinateA CoordinateB CoordinateC

Select the individual coordinates in the vectors for 3D information/transformation. This selector is used for all 3D vectors below, independent if they are position or angle coordinates.

- CoordinateA: The first (X or Theta) coordinate.
- CoordinateB: The second (Y or Phi) coordinate.
- CoordinateC: The third (Z or Rho) coordinate.

23.5 Scan3dCoordinateScale

Name	Scan3dCoordinateScale [Scan3dCoordinateSelector]
Type	FloatReg
Visibility	Expert
Writable in Acquisition Mode	True
AccessMode	RW

Scale factor when transforming a pixel from relative coordinates to world coordinates. A negative scale mirrors the axis. For rectified image axes it is the distance between samples in the rectified image along this axis. The real-world pixel values of the z axis (surface) are internally multiplied with the reciprocal value of Scan3dCoordinateScale[CoordinateC]. The user application has to calculate the real-world pixel value with the calculation: $\text{coordC}[\text{row}, \text{col}] = \text{imageC}[\text{row}, \text{col}] * \text{scaleC} + \text{offsetC}$; Future documentation is in the GenICam SFNC document in chapter 21.4.10 Scan3dCoordinateScale.

23.6 Scan3dCoordinateOffset

Name	Scan3dCoordinateOffset [Scan3dCoordinateSelector]
Type	FloatReg
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW

Offset when transforming a pixel from relative coordinates to world coordinates. This value Scan3dCoordinateOffset[CoordinateC] is internally subtracted from the real-world pixel values of the z axis (surface). The user application has to calculate the real-world pixel value with the calculation: $\text{coordC}[\text{row}, \text{col}] = \text{imageC}[\text{row}, \text{col}] * \text{scaleC} + \text{offsetC}$; The Unit of this feature is given by the Scan3dDistanceUnit feature. Future documentation is in the GenICam SFNC document in chapter 21.4.11 Scan3dCoordinateOffset.

23.7 Scan3dOutputMode

Name	Scan3dOutputMode
Type	Enumeration
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	CalibratedC
	RectifiedC

Controls the Calibration and data organization of the device and the coordinates transmitted.

- CalibratedC: Calibrated 2.5D Depth map. The distance data is expressed in the chosen distance unit. The data is a 2.5D range map. No information on X-Y axes available.
- RectifiedC: Rectified 2.5D depth map. The distance data has been rectified to a uniform sampling pattern in the X and Y direction. The data is a 2.5D range map only.

SENSORCONTROL

Sensor Control Features

24.1 SeqNo

Name	SeqNo
Type	IntReg
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RW

Current selected sequencer number

24.2 BackgroundSuppression

Name	BackgroundSuppression
Type	Enumeration
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	Off On

Sensor background suppression configuration.

- Off: Disable background suppression on sensor
- On: Enable background suppression on sensor

SEQUENCERCONTROL

Category for the Sequencer Control features.

25.1 SequencerMode

Name	SequencerMode
Type	Enumeration
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	Off
	On

Controls if the sequencer mechanism is active.

- Off: Disables the sequencer.
- On: Enables the sequencer.

25.2 SequencerConfigurationMode

Name	SequencerConfigurationMode
Type	Enumeration
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW
EnumEntry	
	Off
	On

Controls if the sequencer configuration mode is active.

- Off: Disables the sequencer configuration mode.
- On: Enables the sequencer configuration mode.

25.3 SequencerFeatureSelector

Name	SequencerFeatureSelector
Type	Enumeration
Visibility	Expert
AccessMode	RW
EnumEntry	
	UserOutputValue
	TargetPosition
	MotionMode
	ScanPosition
	LightBrightness

Select which sequencer feature to control. The feature lists all the features that can be part of a device sequencer set. All the device's sequencer sets have the same features.

- UserOutputValue: Control UserOutputValue feature by the sequencer set.
- TargetPosition: Control TargetPosition feature by the sequencer set.
- MotionMode: Control MotionMode feature by the sequencer set.
- ScanPosition: Control ScanPosition feature by the sequencer set.
- LightBrightness: Control LightBrightness feature by the sequencer set.

25.4 SequencerFeatureEnable

Name	SequencerFeatureEnable [SequencerFeatureSelector]
Type	Boolean
Visibility	Expert
Writable in Acquisition Mode	True
AccessMode	RW

Enable the selected feature and make it active in all the sequencer sets.

25.5 SequencerSetActive

Name	SequencerSetActive
Type	IntReg
Visibility	Expert
AccessMode	RO

Contains the currently active sequencer set.

25.6 SequencerSetStart

Name	SequencerSetStart
Type	Integer
Visibility	Expert
Writable in Acquisition Mode	True
AccessMode	RW
Min	0
Max	7
Inc	1

Sets the initial/start sequencer set, which is the first set used within a sequencer.

25.7 SequencerSetSelector

Name	SequencerSetSelector
Type	Integer
Visibility	Expert
Min	0
Max	7
Inc	1

Select the sequencer set to which further feature settings apply.

25.8 SequencerSetSave

Name	SequencerSetSave [SequencerSetSelector]
Type	Command
Visibility	Expert
Writable in Acquisition Mode	True
AccessMode	WO

Saves the current device state to the sequencer set selected by SequencerSetSelector.

25.9 SequencerSetLoad

Name	SequencerSetLoad [SequencerSetSelector]
Type	Command
Visibility	Expert
Writable in Acquisition Mode	True
AccessMode	WO

Loads the sequencer set selected by SequencerSetSelector in the device. Even if SequencerMode is off, this will change the device state to the configuration of the selected set.

25.10 SequencerPathSelector

Name	SequencerPathSelector [SequencerSetSelector]
Type	Integer
Visibility	Expert
Min	0
Max	3
Inc	1

Select to which branching path further path settings apply.

25.11 SequencerSetNext

Name	SequencerSetNext [SequencerSetSelector] [SequencerPathSelector]
Type	Integer
Visibility	Expert
Writable in Acquisition Mode	True
AccessMode	RW
Min	0
Max	7
Inc	1

Specifies the next sequencer set.

25.12 SequencerTriggerSource

Name	SequencerTriggerSource [SequencerSetSelector] [SequencerPathSelector]
Type	Enumeration
Visibility	Expert
Writable in Acquisition Mode	True
AccessMode	RW
EnumEntry	
	Off
	RTIO4
	DIO1
	DIO2
	DIO3
	DIO4
	DIO5
	DIO6
	DIO7
	DIO8
	DIO10
	DIO11
	DIO12
	DIO13
	DIO14
	DIO15
	DIO16

continues on next page

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DIO17
FrameTrigger
FrameEnd
FieldbusBool0
FieldbusBool1
FieldbusBool2
FieldbusBool3
FieldbusBool4
FieldbusBool5
FieldbusBool6
FieldbusBool7

Specifies the internal signal or physical input line to use as the sequencer trigger source.

- Off: Disables the sequencer trigger.
- RTIO4: Specifies that the trigger will be generated by input RTIO4.
- DIO1: Specifies that the trigger will be generated by input DIO1.
- DIO2: Specifies that the trigger will be generated by input DIO2.
- DIO3: Specifies that the trigger will be generated by input DIO3.
- DIO4: Specifies that the trigger will be generated by input DIO4.
- DIO5: Specifies that the trigger will be generated by input DIO5.
- DIO6: Specifies that the trigger will be generated by input DIO6.
- DIO7: Specifies that the trigger will be generated by input DIO7.
- DIO8: Specifies that the trigger will be generated by input DIO8.
- DIO10: Specifies that the trigger will be generated by input DIO10.
- DIO11: Specifies that the trigger will be generated by input DIO11.
- DIO12: Specifies that the trigger will be generated by input DIO12.
- DIO13: Specifies that the trigger will be generated by input DIO13.
- DIO14: Specifies that the trigger will be generated by input DIO14.
- DIO15: Specifies that the trigger will be generated by input DIO15.
- DIO16: Specifies that the trigger will be generated by input DIO16.
- DIO17: Specifies that the trigger will be generated by input DIO17.
- FrameTrigger: Specifies that the trigger will be generated when the Frame Trigger is received.
- FrameEnd: Specifies that the trigger will be generated by the Frame End signal.
- FieldbusBool0: Specifies that the trigger will be generated by FieldbusBool0.
- FieldbusBool1: Specifies that the trigger will be generated by FieldbusBool1.
- FieldbusBool2: Specifies that the trigger will be generated by FieldbusBool2.
- FieldbusBool3: Specifies that the trigger will be generated by FieldbusBool3.
- FieldbusBool4: Specifies that the trigger will be generated by FieldbusBool4.
- FieldbusBool5: Specifies that the trigger will be generated by FieldbusBool5.
- FieldbusBool6: Specifies that the trigger will be generated by FieldbusBool6.
- FieldbusBool7: Specifies that the trigger will be generated by FieldbusBool7.

25.13 SequencerTriggerActivation

Name	SequencerTriggerActivation [SequencerSetSelector] [SequencerPathSelector]
Type	Enumeration
Visibility	Expert
Writable in Acquisition Mode	True
AccessMode	RW
EnumEntry	
	RisingEdge
	FallingEdge
	AnyEdge

Specifies the mode for the sequencer trigger.

- RisingEdge: Specifies that the trigger is considered valid on the rising edge of the source signal.
- FallingEdge: Specifies that the trigger is considered valid on the falling edge of the source signal.
- AnyEdge: Specifies that the trigger is considered valid on the rising or falling edge of the source signal.

TRANSFERCONTROL

Category that contains the transfer control features.

26.1 TransferStatusSelector

Name	TransferStatusSelector
Type	Enumeration
Visibility	Guru
AccessMode	RW
EnumEntry	Streaming

Select the transfer module status to read using TransferStatus feature.

- Streaming: Data blocks are transmitted when enough data are available.

26.2 TransferStatus

Name	TransferStatus [TransferStatusSelector]
Type	Boolean
Visibility	Guru
AccessMode	RO

Reads the status of the Transfer module signal selected by TransferStatusSelector.

26.3 TransferSelector

Name	TransferSelector
Type	Enumeration
Visibility	Expert
AccessMode	RO
EnumEntry	All

Select which stream transfers are currently controlled by the selected Transfer features.

- All: The transfer features control all the data streams simultaneously.

26.4 TransferQueueMaxBlockCount

Name	TransferQueueMaxBlockCount [TransferSelector]
Type	Integer
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW
Min	1
Max	127
Inc	1

Controls the maximum number of data blocks that can be stored in the block queue of the selected stream. A value of **1** (default) disables parallel acquisition during transmission, resulting in a sequential process of image recording, processing and transmission. Values **>1** define the count of camera-buffered data blocks containing processed image data ready for transfer. **TransferQueueMaxBlockCount > 1** implicitly enables one raw image buffer as well. This means, that the device is able to record new data (recordingActive) before an empty data block is available. As soon as a free data block is available, the processing is started and the data block filled.

TRANSPORTLAYERCONTROL

Category that contains the transport Layer control features.

27.1 PayloadSize

Name	PayloadSize
Type	IntReg
Visibility	Expert
AccessMode	RO
Unit	B

Provides the number of bytes transferred for each image or chunk on the stream channel. This includes any end-of-line, end-of-frame statistics or other stamp data. This is the total size of data payload for a data block.

27.2 GenDCStreamingMode

Name	GenDCStreamingMode
Type	Enumeration
Visibility	Guru
AccessMode	RW
EnumEntry	Off
	On

Controls the device's streaming format. Note: This feature is meant to globally control the data streaming format of the device. As such, it has priority over other features (except TestPayloadFormatMode) that might influence the output payload format or content

- Off: The device will only stream data in its native format.
- On: The device will stream all its data in the generic GenDC format.

27.3 GenDCStreamingStatus

Name	GenDCStreamingStatus
Type	Enumeration
Visibility	Guru
AccessMode	RO
EnumEntry	
	Off
	On

Returns whether the current device data streaming format is GenDC. This value is conditioned by the GenDCStreamingMode. This feature can be used to determine if the device will stream in its native or in the generic GenDC format and if GenDC related features like GenDCDescriptor and GenDCFlowmappingTable can be used.

- Off: The device will only stream data in its native format.
- On: The device will stream all its data in the generic GenDC format.

USERSETCONTROL

Category that contains the user set control features.

28.1 UserSetSelector

Name	UserSetSelector
Type	Enumeration
Visibility	Beginner
AccessMode	RW
EnumEntry	
	Default
	UserSet0
	UserSet1

Select the feature User Set to load or save

- Default: Select the factory setting user set.
- UserSet0: Select the user set 0.
- UserSet1: Select the user set 1.

28.2 UserSetLoad

Name	UserSetLoad [UserSetSelector]
Type	Command
Visibility	Beginner
Writable in Acquisition Mode	False
AccessMode	WO

Load the User Set specified by UserSetSelector to the device and makes it active.

28.3 UserSetSave

Name	UserSetSave [UserSetSelector]
Type	Command
Visibility	Beginner
Writable in Acquisition Mode	False
AccessMode	WO

Save the User Set specified by UserSetSelector to the non-volatile memory of the device.

28.4 UserSetDefault

Name	UserSetDefault
Type	Enumeration
Visibility	Beginner
AccessMode	RW
EnumEntry	
	Default
	UserSet0
	UserSet1

Select the feature User Set to load and make active by default when the device is reset.

- Default: Select the factory setting user set.
- UserSet0: Select the user set 0.
- UserSet1: Select the user set 1.

WLICONTROL

WLI configuration features

29.1 NominalWavelength

Name	NominalWavelength
Type	FloatReg
Visibility	Expert
AccessMode	RO

Nominal wavelength in nm of the device.

29.2 WavelengthCorrection

Name	WavelengthCorrection
Type	FloatReg
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RW

Correction factor of the nominal wavelength. The effective wavelength depends on the optics and could be corrected with this parameter.

29.3 EffectiveWavelength

Name	EffectiveWavelength
Type	FloatReg
Visibility	Expert
AccessMode	RO

Effective wavelength in nm used for parameter calculation.

29.4 ScanSpeed

Name	ScanSpeed
Type	Float
Visibility	Beginner
Operation Mode	Topography Tomography
Writable in Acquisition Mode	False
AccessMode	RW
Min	0.1
Max	depends on feature <i>SpeedMax</i>

Configure the speed used for measurements [mm/s]. This configuration change also the sensor configuration.

29.5 TargetVerticalSpacing

Name	TargetVerticalSpacing
Type	FloatReg
Visibility	Beginner
Operation Mode	Topography Tomography
Writable in Acquisition Mode	False
AccessMode	RW

Configure the target vertical spacing [um] and trigger recalculation of sensor parameters. The true vertical spacing could be read back in ActualVerticalSpacing.

29.6 ActualVerticalSpacing

Name	ActualVerticalSpacing
Type	FloatReg
Visibility	Beginner
Operation Mode	Topography Tomography
Writable in Acquisition Mode	False
AccessMode	RO

Current vertical spacing [um]. This feature returns the true vertical spacing. The true Vertical spacing is as close as possible to TargetVerticalSpacing.

29.7 TargetVerticalSpreading

Name	TargetVerticalSpreading
Type	Float
Visibility	Expert
Operation Mode	Topography Tomography
Writable in Acquisition Mode	False
AccessMode	RW
Unit	%
Min	0.0
Max	100.0

This feature configures the target camera exposure spreading over the whole frame. With a spreading of 100.0 %, the set total exposure time of the camera is maximally distributed over the entire duration of a frame. A spread of 0.0% concentrates the total exposure as compactly as possible in time (and therefore also in z-distance in the case of WLI). The configurable ExposureRatio determines the effectively exposed portion of the maximum possible exposure time and therefore also how much leeway there is when distributing the time over a frame. Vertical spreading is subordinate to the exposure ratio and never changes it. Its effective value can be read by means of the feature ActualVerticalSpreading.

29.8 ActualVerticalSpreading

Name	ActualVerticalSpreading
Type	FloatReg
Visibility	Expert
Operation Mode	Topography Tomography
Writable in Acquisition Mode	False
AccessMode	RO
Unit	%

This feature returns the effective vertical spreading when DeviceOperationMode=Topography or Tomography. The target value is configured using TargetVerticalSpreading.

29.9 ScanRange

Name	ScanRange [VerticalSegmentSelector]
Type	Float
Visibility	Beginner
Operation Mode	Topography Tomography
Writable in Acquisition Mode	False
AccessMode	RW
Min	0.01
Inc	0.001
Max	depends on feature <i>ScanRangeMax</i>

Configure the target scan range [mm]. And read back the actual scan range.

29.10 TargetScanRange

Name	TargetScanRange [VerticalSegmentSelector]
Type	Float
Visibility	Beginner
Operation Mode	Topography Tomography
Writable in Acquisition Mode	False
AccessMode	RW
Min	0.01
Inc	0.001
Max	depends on feature <i>ScanRangeMax</i>

Configure the target scan range [mm]. The current value could be read by the ActualScanRange feature.

29.11 ScanRangeMax

Name	ScanRangeMax [VerticalSegmentSelector]
Type	FloatReg
Visibility	Invisible
AccessMode	RO

Internal used feature to expose the ScanRange limits. The maximum possible value for the ScanRange depends on the ScanPosition, AccelerationRamp and the stage limits.

29.12 ActualScanRange

Name	ActualScanRange [VerticalSegmentSelector]
Type	FloatReg
Visibility	Beginner
Operation Mode	Topography Tomography
Writable in Acquisition Mode	False
AccessMode	RO

Current scan range [mm]. This feature return the true scan range. The true scan range is as close as possible to TargetScanRange.

29.13 ExposureRatio

Name	ExposureRatio
Type	Float
Visibility	Beginner
Operation Mode	Topography Tomography
Writable in Acquisition Mode	False
AccessMode	RW
Min	0.0
Max	1.0

Configure the camera's exposure ratio (0.0 - 1.0). The ratio defines how long the exposure is on during a measurement. Default is 1.0.

29.14 RecordingNFrames

Name	RecordingNFrames [VerticalSegmentSelector]
Type	IntReg
Visibility	Guru
Operation Mode	Topography Tomography
Writable in Acquisition Mode	False
AccessMode	RW

Get the number of recorded frames. The value change dependent on the configured ScanRange and the VerticalSpacing. It is only writable in the LowLevel mode.

29.15 PixelDimension

Name	PixelDimension
Type	FloatReg
Visibility	Guru
AccessMode	RO

This feature is Deprecated!

Deprecated! Use SensorPixelWidth and SensorPixelHeight. Return the physical size of a pixel in um on the image sensor.

29.16 LateralResolution

Name	LateralResolution
Type	FloatReg
Visibility	Guru
AccessMode	RO
Unit	um

Return the optical resolution in um depending on the magnification.

29.17 FrameRate

Name	FrameRate
Type	FloatReg
Visibility	Guru
AccessMode	RO

Recording frame rate resulting with the current camera settings [fps]. If an external reference trigger signal is fed to the camera, then this determines the effective frame rate. In this case, 'FrameRate' yields the largest possible frame

rate that an external trigger signal can produce with given settings. Hence, the return value is suitable to check for frame rate violations. However, it does not generally reflect the actual frame rate with an external reference trigger signal.

29.18 VerticalSegmentSelector

Name	VerticalSegmentSelector
Type	Enumeration
Visibility	Beginner
Operation Mode	Topography Tomography
AccessMode	RW
EnumEntry	
	Segment0
	Segment1
	Segment2
	Segment3

Select the Vertical Segment which is configured.

- Segment0: Configuration of Segment0.
- Segment1: Configuration of Segment1.
- Segment2: Configuration of Segment2.
- Segment3: Configuration of Segment3.

29.19 VerticalSegmentEnable

Name	VerticalSegmentEnable [VerticalSegmentSelector]
Type	Boolean
Visibility	Expert
Operation Mode	Topography Tomography
Writable in Acquisition Mode	False
AccessMode	RW

Controls if the selected vertical segment is active. True: Selected segment is enabled. False: Selected segment is disabled.

29.20 ScanPosition

Name	ScanPosition [VerticalSegmentSelector]
Type	Float
Visibility	Beginner
Operation Mode	Topography Tomography
Writable in Acquisition Mode	True
AccessMode	RW
Inc	0.001
Min	depends on feature <i>ScanPositionMin</i>
Max	depends on feature <i>ScanPositionMax</i>

Configure the scan position [mm]. The configured position is the center position of the scan range. The AccelerationRamp is used for acceleration and deceleration at the top and bottom of the scan range.

29.21 ScanPositionMin

Name	ScanPositionMin [VerticalSegmentSelector]
Type	FloatReg
Visibility	Invisible
AccessMode	RO

Internal feature used to expose the ScanPosition limits. The minimum possible value for the ScanPosition depends on the configured ScanRange, AccelerationRamp and the stage limits.

29.22 ScanPositionMax

Name	ScanPositionMax [VerticalSegmentSelector]
Type	FloatReg
Visibility	Invisible
AccessMode	RO

Internal feature used to expose the ScanPosition limits. The maximum possible value for the ScanPosition depends on the configured ScanRange, AccelerationRamp and the stage limits.

29.23 OverallScanPosition

Name	OverallScanPosition
Type	FloatReg
Visibility	Expert
Writable in Acquisition Mode	False
AccessMode	RO

Configure the scan position [mm]. The configured position is the center position of the scan range. The AccelerationRamp is used for acceleration and deceleration at the top and bottom of the scan range.

29.24 OverallScanRange

Name	OverallScanRange
Type	FloatReg
Visibility	Expert
Operation Mode	Topography Tomography
Writable in Acquisition Mode	False
AccessMode	RO

Configure the scan range [mm]. The AccelerationRamp is used for acceleration and deceleration at the top and bottom of the scan range.

29.25 OverallRecordingNFrames

Name	OverallRecordingNFrames
Type	IntReg
Visibility	Guru
Writable in Acquisition Mode	False
AccessMode	RO

Get the number of recorded frames. The value change dependent on the configured ScanRange and the VerticalSpacing. It is only writable in the LowLevel mode.

29.26 SegmentedVolumeActive

Name	SegmentedVolumeActive
Type	Boolean
Visibility	Expert
Operation Mode	Topography Tomography
AccessMode	RO

Indicates if multiple vertical segments are enabled.

30.1 Event And Status Diagram

The gen4 devices contain three sequence control loops running in parallel:

- **Acquisition:** Main Loop. Controls the measurement sequence and synchronizes the other control loops. State information is available through the **AcquisitionStatus** features.
- **Transfer:** Controls the transmission of the measurement data. State information is available through the **TransferStatus** features.
- **Stage:** Controls the motion. State information is available through the **StageStatus** features.

The loops can be monitored and controlled through features (Software) or digital I/Os (Hardware).

Table 30.1: Monitoring and Control Options

Type	Description	Hardware	Software
Status	Exposes the current status of a control loop	Digital Output (level)	AcquisitionStatus TransferStatus StageStatus
Events	Indicate the state transition of a control loop		Events CounterEventSource SequencerTrigger-Source
Triggers	Initiate a state transition. Based on the device configuration, a transition can be triggered either explicitly or automatically after the success of a previous state.	Digital Input (edge)	TriggerSoftware

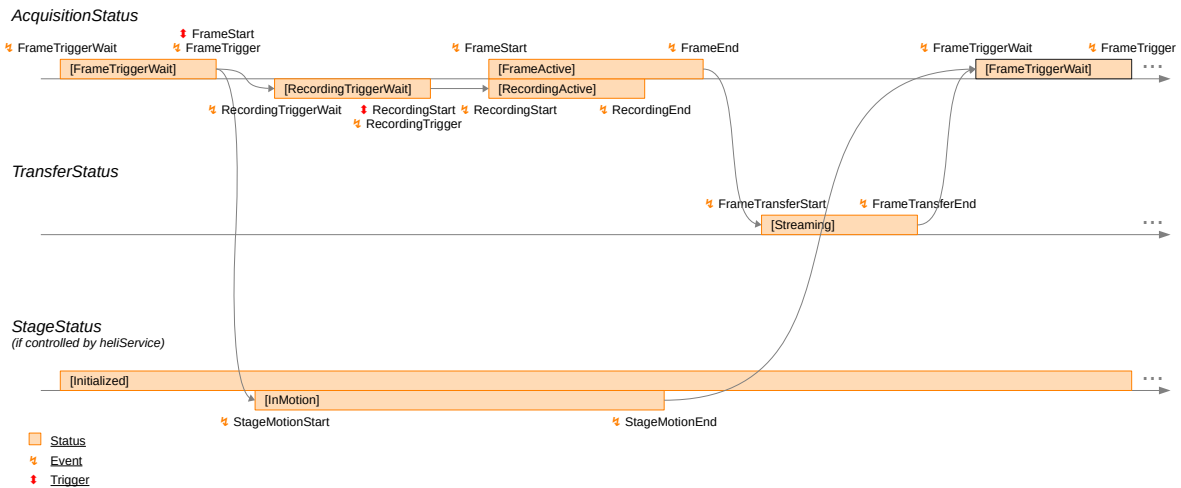


Fig. 30.1: Default “Event And Status Diagram” - single Image Buffer on the device

To reduce the overall cycle time, multiple buffers can be enabled on the device. Using values >1 for the **Transfer-QueueMaxBlockCount** feature allocates additional buffers

on the device so that an acquisition during data transfer is possible.

Therefore, the *Event And Status Diagram* below applies.

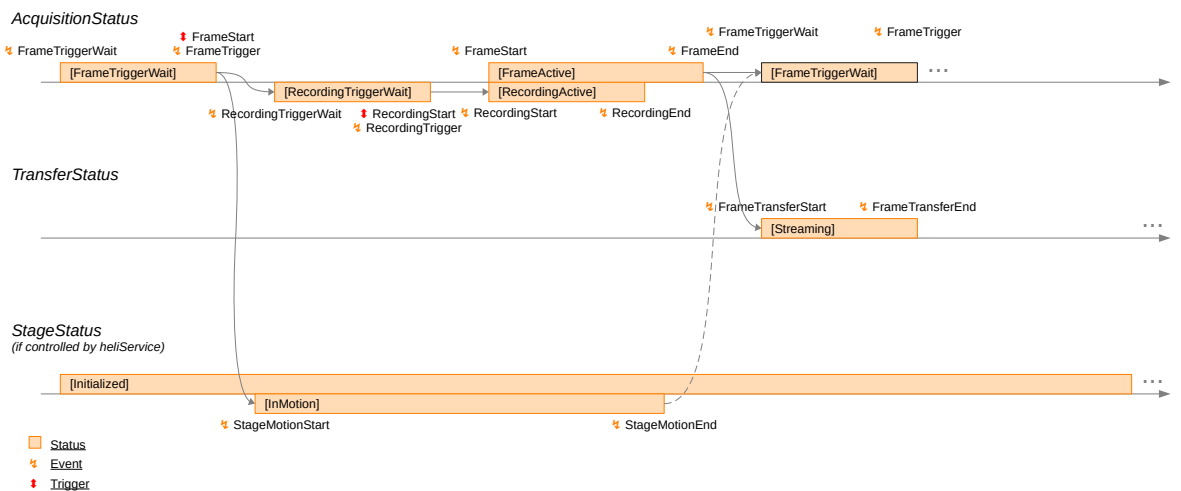


Fig. 30.2: “Event And Status Diagram” with multiple buffers enabled on the device